

**CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY**

COLLEGE OF COMPUTER STUDIES

Software Requirements Specifications
for
TREVORA

Change History

Version	Date	Description	Author(s)
1.0	14 May 2026	Initial SRS draft for Trevora	Lauro, Jhon Gil V. Unabia, Brent Jelson D. Gamallo, Benz Leo Sy, Jullian Philip S. Potane, Kate Marielle C.

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1. Introduction

1.1. Purpose

The purpose of this Software Requirements Specification is to define the requirements of Trevora, a web-based vehicle service history system for service understanding and mechanic handoff. This document describes the system scope, user roles, functional requirements, non-functional requirements, use cases, diagrams, constraints, assumptions, risks, and validation considerations. It serves as a guide for the development team, project adviser, and stakeholders to ensure that the system follows the approved proposal objectives.

1.2. Scope

Trevora is a web-based vehicle service history system designed for vehicle owners. The system helps vehicle owners capture, validate, consolidate, understand, and share vehicle maintenance and repair records.

The system allows vehicle owners to submit service records through receipt image upload, voice input, or manual entry. The system extracts or captures key service details such as service date, service type, parts replaced, shop/mechanic name, cost, and remarks. Before saving, the owner can review, correct, and confirm the service details.

Validated records are stored under the correct registered vehicle profile and organized into one chronological service history. The system also shows AI-generated explanations beside service records so vehicle owners can understand what was done and why it matters.

For mechanic handoff, the owner can generate a one-time QR access request for the service history of the selected registered vehicle. The mechanic can request access, but the owner must approve the request first. Once approved, the mechanic can view the approved vehicle service history through temporary read-only access and use AI-assisted search to find related past service information. Access automatically expires after a system-set time limit.

The MVP does not include dealership integration, manufacturer database integration, vehicle diagnostic hardware, predictive maintenance, legal vehicle ownership transfer, or mechanic editing of owner records.

1.3. Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification.
Trevora	The official project name for the vehicle service history system.
Vehicle Owner	The primary user who submits, validates, views, manages, and shares the service history of their registered vehicle.
Vehicle Profile	A system record that stores information about a registered vehicle and links service records to that vehicle.

Service Record	A structured record of maintenance or repair activity.
Receipt Image	A submitted image of a service receipt used as an input source for extracting service details.
Voice Input	Spoken service information submitted by the user and converted into text-based service data.
Manual Entry	A backup input method where the vehicle owner directly enters service details into system fields.
Validated Service Record	A service record reviewed, corrected, and confirmed by the owner before storage.
QR Access Request	A one-time QR code generated by the vehicle owner to allow a mechanic to request temporary access to the service history of the selected registered vehicle.
Temporary Read-Only Access	Limited mechanic access through a temporary page where the service history of the approved vehicle can be viewed but not edited or deleted, and access expires after a system-set time limit.
AI-Assisted Search	A feature that helps mechanics find relevant approved records using natural-language search.
Use Case	A user-centered transaction or system function required by the SRS.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
AI-Generated Explanation	A simple explanation shown beside a service record to help the vehicle owner understand what was done and why it matters.
Owner Approval	The required confirmation from the vehicle owner before a mechanic can view shared service records.
System-Set Time Limit	The fixed access duration set by the system for temporary mechanic access.
Mechanic / Service Personnel	A secondary user who can request temporary read-only access to the service history of a vehicle approved by the vehicle owner.

1.4. References

The following references support the system requirements related to vehicle service records, maintenance data quality, receipt image extraction, speech-to-text input, AI-generated explanations, QR security, and service history sharing.

- **Chen, J., Ruan, Y., Guo, L., & Lu, H. (2020).** BCVehis: A blockchain-based service prototype of vehicle history tracking for used-car trades in China. *IEEE Access*, 8, 214842–214851.
Use for: vehicle history tracking, traceable records, service history sharing.

- **Dua, M., Akanksha, & Dua, S. (2023).** Noise robust automatic speech recognition: Review and analysis. *International Journal of Speech Technology*, 26(2), 475–519.
Use for: speech-to-text / voice input.
- **Focardi, R., Luccio, F. L., & Wahsheh, H. A. M. (2019).** Usable security for QR code. *Journal of Information Security and Applications*, 48, 102369.
Use for: QR access, QR security, one-time access concerns.
- **Ha, H. T., & Horak, A. (2022).** Information extraction from scanned invoice images using text analysis and layout features. *Signal Processing: Image Communication*, 102, 116601.
Use for: receipt image extraction / OCR.
- **Hemphill, T. A., Longstreet, P., & Banerjee, S. (2022).** Automotive repairs, data accessibility, and privacy and security challenges: A stakeholder analysis and proposed policy solutions. *Technology in Society*, 71, 102090.
Use for: automotive repair data access, privacy, sharing issues.
- **Hodkiewicz, M., & Ho, M. T. (2016).** Cleaning historical maintenance work order data for reliability analysis. *Journal of Quality in Maintenance Engineering*, 22(2), 146–163.
Use for: cleaning maintenance records, reliability of historical records.
- **Hu, Z., Zhang, X., & Xiong, H. (2025).** A topic joint model for knowledge extraction from unstructured maintenance records. *Engineering Applications of Artificial Intelligence*, 142, 109743.
Use for: AI/knowledge extraction from unstructured maintenance records.
- **Liang, J. S. (2020).** A process-based automotive troubleshooting service and knowledge management system in collaborative environment. *Robotics and Computer-Integrated Manufacturing*, 61, 101836.
Use for: automotive service knowledge management.
- **Lin, C.-J., Liu, Y.-C., & Lee, C.-L. (2022).** Automatic receipt recognition system based on artificial intelligence technology. *Applied Sciences*, 12(2), 853.
Use for: AI receipt recognition.
- **Madhikermi, M., Kubler, S., Robert, J., Buda, A., & Framling, K. (2016).** Data quality assessment of maintenance reporting procedures. *Expert Systems with Applications*, 63, 145–164.
Use for: maintenance data quality and reporting reliability.
- **Rong, Y., Leemann, T., Nguyen, T.-T., Fiedler, L., Qian, P., Unhelkar, V., Seidel, T., Kasneci, G., & Kasneci, E. (2024).** Towards human-centered explainable AI: A survey of user studies for model explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(4), 2104–2122.
Use for: AI-generated explanations / user understanding.
- **Sala, R., Bertoni, M., Pirola, F., & Pezzotta, G. (2023).** Improvement of maintenance-based product-service system offering through field data: A case study. *Production & Manufacturing Research*, 11(1).
Use for: using field maintenance data and service improvement.
- **Sala, R., Pirola, F., Pezzotta, G., & Cavalieri, S. (2022).** Data-driven decision making in maintenance service delivery process: A case study. *Applied Sciences*, 12(15), 7395.
Use for: maintenance service data and decision-making.
- **Wang, R. Y., & Strong, D. M. (1996).** Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems*, 12(4), 5–33.
Use for: data quality, completeness, accessibility, usefulness.
- **Webster, B. M., Hare, C. E., & McLeod, J. (1999).** Records management practices in small and medium-sized enterprises: A study in North-East England. *Journal of Information Science*, 25(4), 283–294.
Use for: records management and documentation practices.

2. Overall Description

2.1. Product perspective

Trevora is an independent web-based system for managing vehicle service history. It is designed mainly for vehicle owners who collect service records from different repair visits, shops, receipts, voice notes, or manual entries. The system does not require repair shops, dealerships, or manufacturers to use the platform.

The system has four main modules: Service Record Input, Service Data Validation and Correction, Unified Vehicle Service History Consolidation, and AI-Assisted Service Understanding and Mechanic Handoff. These modules help owners capture service information, check details before saving, organize records under the correct vehicle, understand service records through AI-generated explanations, and share service history with mechanics through owner-approved temporary access.

Trevora does not connect to vehicle diagnostic hardware, dealership systems, manufacturer databases, or third-party repair platforms in the MVP. All service history information depends on the records submitted, reviewed, and confirmed by the vehicle owner.

2.2. User characteristics

User Type	Description	Primary System Activities
Vehicle Owner	A user who owns or manages one or more vehicles.	Upload receipt images, record voice input, enter service details manually, review and correct service details, save validated records, view service history, read AI-generated explanations, generate QR access requests, and approve or deny mechanic access.
Mechanic / Service Personnel	A secondary user who can request temporary access to a vehicle's service history after owner approval.	Scan QR code, request access, wait for owner approval, view approved vehicle service history in read-only mode, and use AI-assisted search on shared records.
Administrator	A user responsible for system oversight and account support.	Maintain accounts, monitor system access, and support data integrity controls.

2.3. Constraints

- The system shall be accessible through a modern web browser.
- The system shall require internet connection for account access, cloud storage, OCR, speech-to-text, AI-generated explanations, AI-assisted record search, and QR-based mechanic access.
- The system shall not diagnose vehicle problems or replace mechanic judgment.
- The system shall not allow mechanics to create, edit, or delete owner service records.
- The system shall only allow mechanics to view the service history of the registered vehicle approved by the vehicle owner.
- The QR access request shall be one-time use only.

- Temporary mechanic access shall automatically expire after the system-set time limit.
- The system shall not connect to dealership systems, manufacturer databases, third-party repair platforms, or vehicle diagnostic hardware in the MVP.

2.4. Assumptions and dependencies

- Vehicle owners are expected to have access to an internet-enabled device and a modern web browser.
- Receipt image extraction depends on the clarity, quality, and format of the uploaded receipt image.
- Speech-to-text accuracy depends on the clarity of the audio input and background noise.
- AI-generated explanations depend on the available details inside validated service records.
- AI-assisted mechanic search depends on the service history approved by the vehicle owner.
- Vehicle owners are responsible for reviewing, correcting, and confirming service details before saving.
- Mechanics can only access a vehicle's service history after owner approval.
- If OCR, speech-to-text, or AI services are unavailable, the system shall still allow manual entry, record viewing, and basic service history access when possible.

Risk	Impact	Mitigation
Poor receipt image quality	Extracted fields may be inaccurate or incomplete.	The system will show editable fields and missing-field warnings.
Unclear voice input	Transcribed service details may be incorrect.	The system will require user review and correction before saving.
Wrong or incomplete saved record	The service history may become unreliable.	The system will require validation and owner confirmation before saving.
Incorrect AI-generated explanation	The owner may misunderstand the service record.	The system will show the original service record beside the AI explanation.
AI search returns unrelated records	The mechanic may receive wrong or less useful results.	The system will search only approved vehicle service history and show original record details.
Unapproved mechanic access	Vehicle service history privacy may be affected.	The system will require owner approval before mechanic access.
QR code reuse	Shared records may be accessed again without permission.	The QR code will be one-time use only.
Expired access still visible	The mechanic may view records longer than allowed.	The system will automatically expire access after the system-set time limit.
Internet connection problem	AI features, QR access, or cloud storage may not work properly.	The system will allow manual entry and existing record viewing when possible.

3. Specific Requirements

3.1. External interface requirements

3.1.1. Hardware interfaces

The system shall be accessible through computers, tablets, and smartphones that support a modern web browser. Receipt image upload requires a device camera or file upload capability. Voice input requires a working microphone. Mechanics may use their own smartphone, tablet, or computer to scan the QR code and access approved service history.

3.1.2. Software interfaces

The system shall run through a modern web browser. It may use OCR services for receipt image extraction, speech-to-text services for voice input, AI services for service record explanations, and AI-assisted search for mechanic handoff. The system shall store and retrieve vehicle profiles, service records, QR access requests, and mechanic access status through its database.

3.1.3. Communications interfaces

The system shall use secure HTTPS connections for user login, service record storage, receipt image processing, voice input processing, AI-generated explanations, QR access requests, owner approval, temporary mechanic access, and AI-assisted record search.

3.2. Functional requirements

Each module below maps to one approved general objective. Each use case represents a transaction or system function derived from the related specific objective.

Objective ID	General Objective	Module Name	Module Purpose
GO-1	To improve service record input by allowing vehicle owners to create or select a registered vehicle profile before submitting maintenance and repair information through receipt image upload, voice input, or manual entry, achieving at least 85% field extraction accuracy for receipt-based input and at least 85% transcription accuracy for voice-based input.	Service Record Input	Capture service data from receipt image upload, voice input, and manual entry in one structured entry format.
GO-2	To improve the reliability of extracted vehicle service data by allowing users to review, validate, and correct extracted service details before saving, reducing incorrect or incomplete service entries by at least 30%.	Service Data Validation and Correction	Ensure captured records are reviewed, corrected, and confirmed before storage.
GO-3	To reduce fragmented vehicle service data by consolidating validated maintenance and	Unified Vehicle Service History Consolidation	Store and display validated service records

	repair records into a unified chronological service history for each registered vehicle, ensuring 100% of approved service entries are stored under the correct vehicle profile.		under the correct vehicle profile in a centralized chronological history.
GO-4	To improve the use of vehicle service history by providing AI-generated explanations for vehicle owners and QR-based AI record search for mechanics, reducing the time needed to find or understand relevant service information by at least 30%.	AI-Assisted Service Understanding and Mechanic Handoff	Provide AI-generated service explanations and temporary owner-approved read-only mechanic access with AI-assisted record search.

Module 1: Service Record Input

General Objective 1: To improve service record input by allowing vehicle owners to create or select a registered vehicle profile before submitting maintenance and repair information through receipt image upload, voice input, or manual entry, achieving at least 85% field extraction accuracy for receipt-based input and at least 85% transcription accuracy for voice-based input.

Specific Objective-to-Requirement Mapping

Specific Objective	Use Case / Function	Requirement ID
1.1	UC-EXT-01 Create or Select Registered Vehicle Profile	FR-EXT-01
1.2	UC-EXT-02 Upload Receipt and Extract Details	FR-EXT-02
1.3	UC-EXT-03 Record Voice Service Information	FR-EXT-03
1.4	UC-EXT-04 Enter Service Details Manually	FR-EXT-04
1.5	UC-EXT-05 Convert Input to Structured Service Entry	FR-EXT-05
Requirement ID	Requirement Statement	
FR-EXT-01	The system shall allow vehicle owners to create or select a registered vehicle profile before submitting a service record.	
FR-EXT-02	The system shall extract key service details from receipt images, including service date, service type, parts replaced, shop/mechanic name, cost, and remarks.	
FR-EXT-03	The system shall convert spoken service information into text-based service data using speech-to-text input.	
FR-EXT-04	The system shall provide manual entry fields for required and relevant service record data when receipt image upload or voice input is unavailable.	

FR-EXT-05	The system shall convert receipt image, voice, or manual input data into a structured service entry format containing the required service record fields, including vehicle profile, service date, service type, cost, and other relevant service details.
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UC-EXT-01 Create or Select Registered Vehicle Profile

Use Case Item	Description
Use Case Name	Create or Select Registered Vehicle Profile
Related Specific Objective	1.1
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in.
Trigger	The vehicle owner wants to submit a service record.
Main Flow	The owner opens the service record input module. The system checks whether the owner has an existing registered vehicle profile. If a vehicle profile exists, the owner selects the vehicle that the service record belongs to. If no vehicle profile exists, the owner creates a new vehicle profile by entering vehicle details such as make, model, year, nickname, plate number, VIN/chassis number, and odometer. The system saves or selects the vehicle profile before allowing the owner to continue with service record input.
Postcondition	A registered vehicle profile is selected and ready to be linked to the service record.
Alternative Flow	If the owner has multiple registered vehicles, the system allows the owner to switch or select the correct vehicle before submitting the service record.
Exception Flow	If the vehicle profile cannot be created or loaded, the system displays an error message and prevents service record submission until a valid vehicle profile is selected or created.
Requirement ID	FR-EXT-01

Use Case Diagram - UC-EXT-01

Create or Select Registered Vehicle Profile

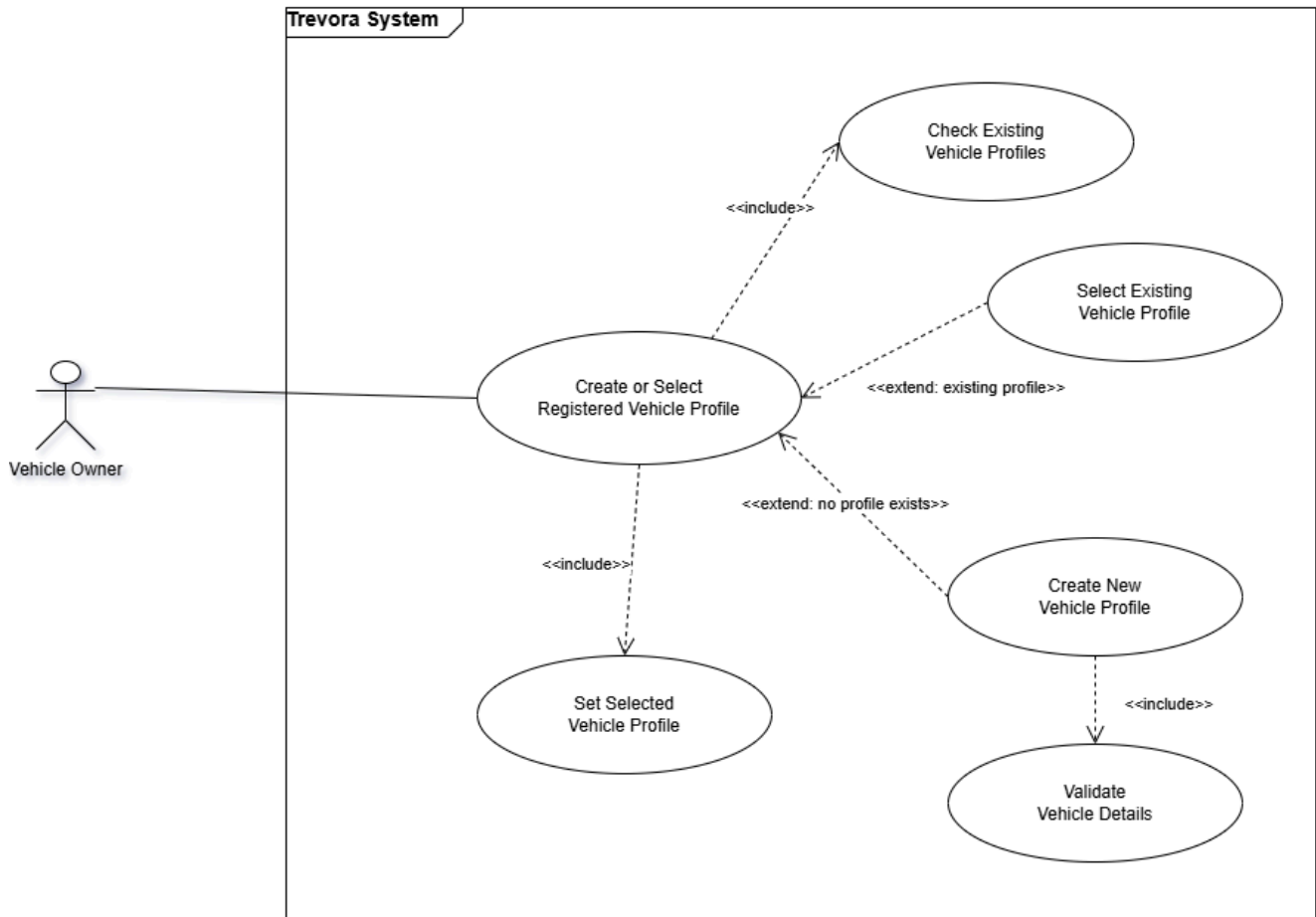


Figure - Use Case Diagram for UC-EXT-01 Create or Select Registered Vehicle Profile

Activity Diagram - UC-EXT-01

Create or Select Registered Vehicle Profile

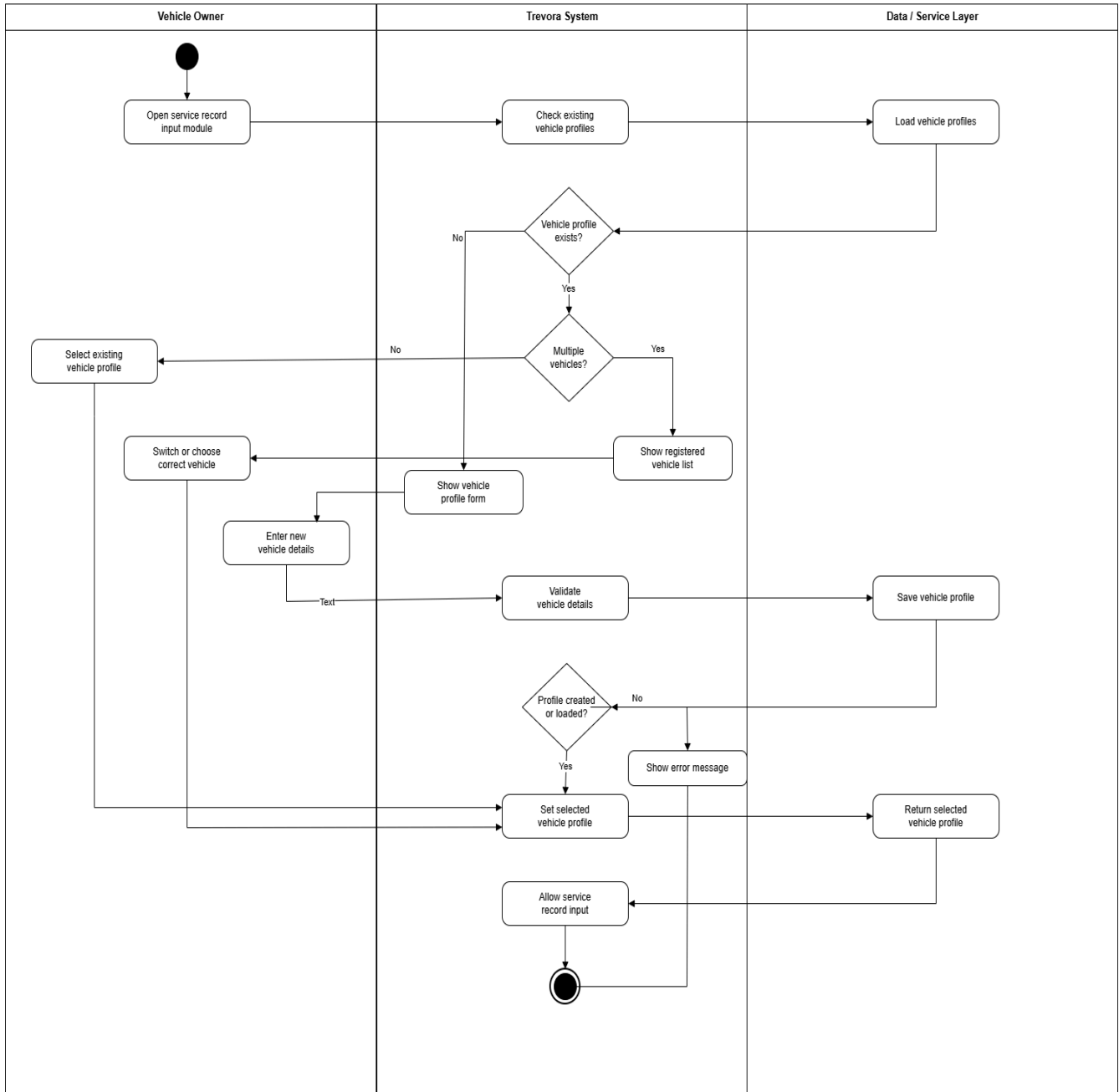


Figure - Activity Diagram for UC-EXT-01 Create or Select Registered Vehicle Profile

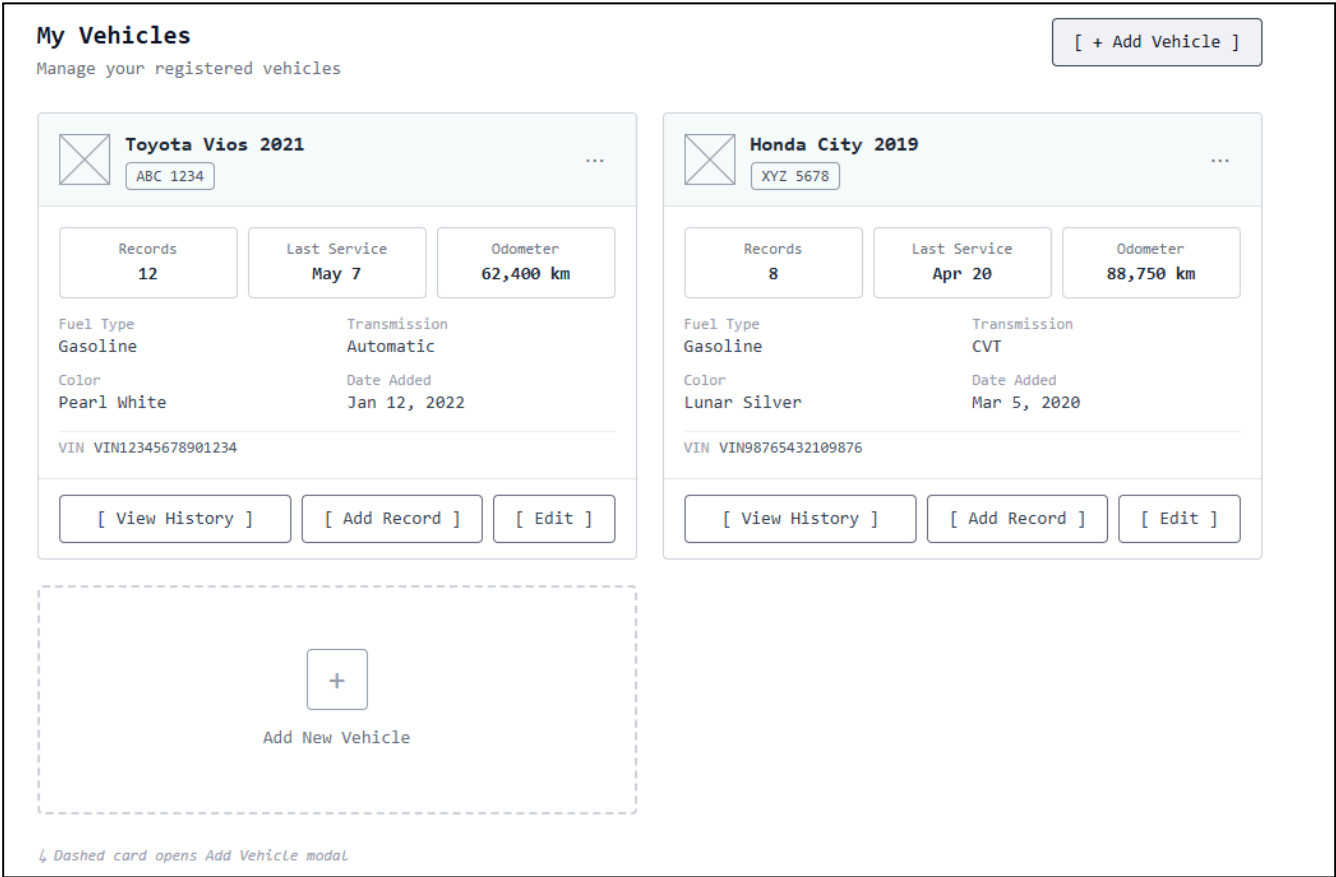


Figure - Low-Fidelity Wireframe for UC-EXT-01 Create or Select Registered Vehicle Profile

UC-EXT-02 Upload Receipt and Extract Details

Use Case Item	Description
Use Case Name	Upload Receipt and Extract Details
Related Specific Objective	1.2
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has selected or created a registered vehicle profile.
Trigger	The vehicle owner chooses receipt image upload as the service record input method.
Main Flow	The owner uploads a receipt image or captures one using the device camera. The system checks the image quality and informs the owner if the image may affect extraction accuracy. The system processes the receipt image using OCR and AI-assisted field mapping. The system extracts key service details such as service date, service type, parts replaced, shop/mechanic name, cost, and remarks. The extracted details are prepared as an AI-assisted service draft.

Postcondition	Extracted service details are displayed in an editable structured draft for review.
Alternative Flow	If some details are missing, unclear, or unreadable, the system leaves the affected fields blank or marks them for user review.
Exception Flow	If the image upload fails, the file format is not supported, or the image cannot be processed, the system displays an error message and allows the owner to upload another image or use manual entry.
Requirement ID	FR-EXT-02

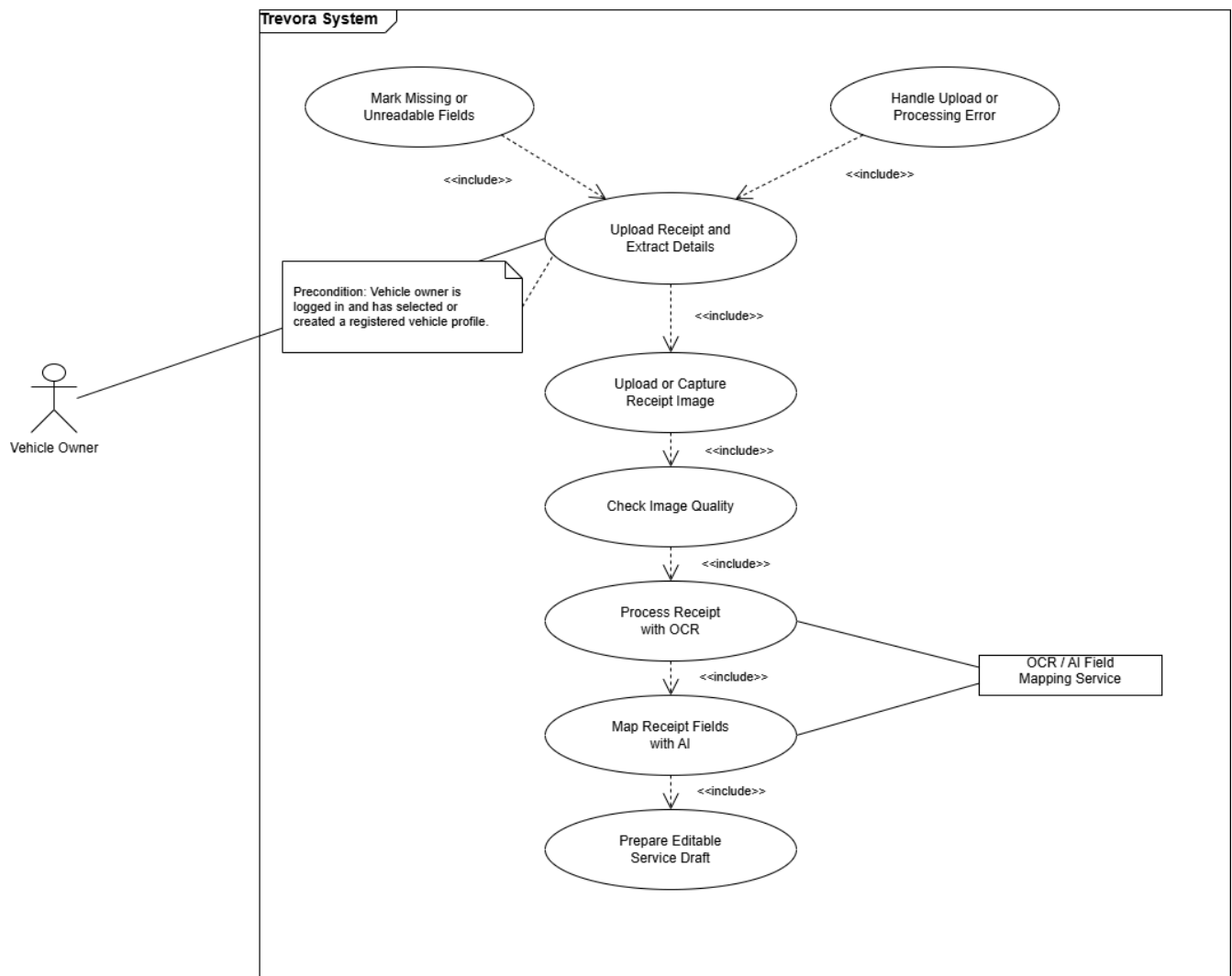


Figure - Use Case Diagram for UC-EXT-02 Upload Receipt and Extract Details

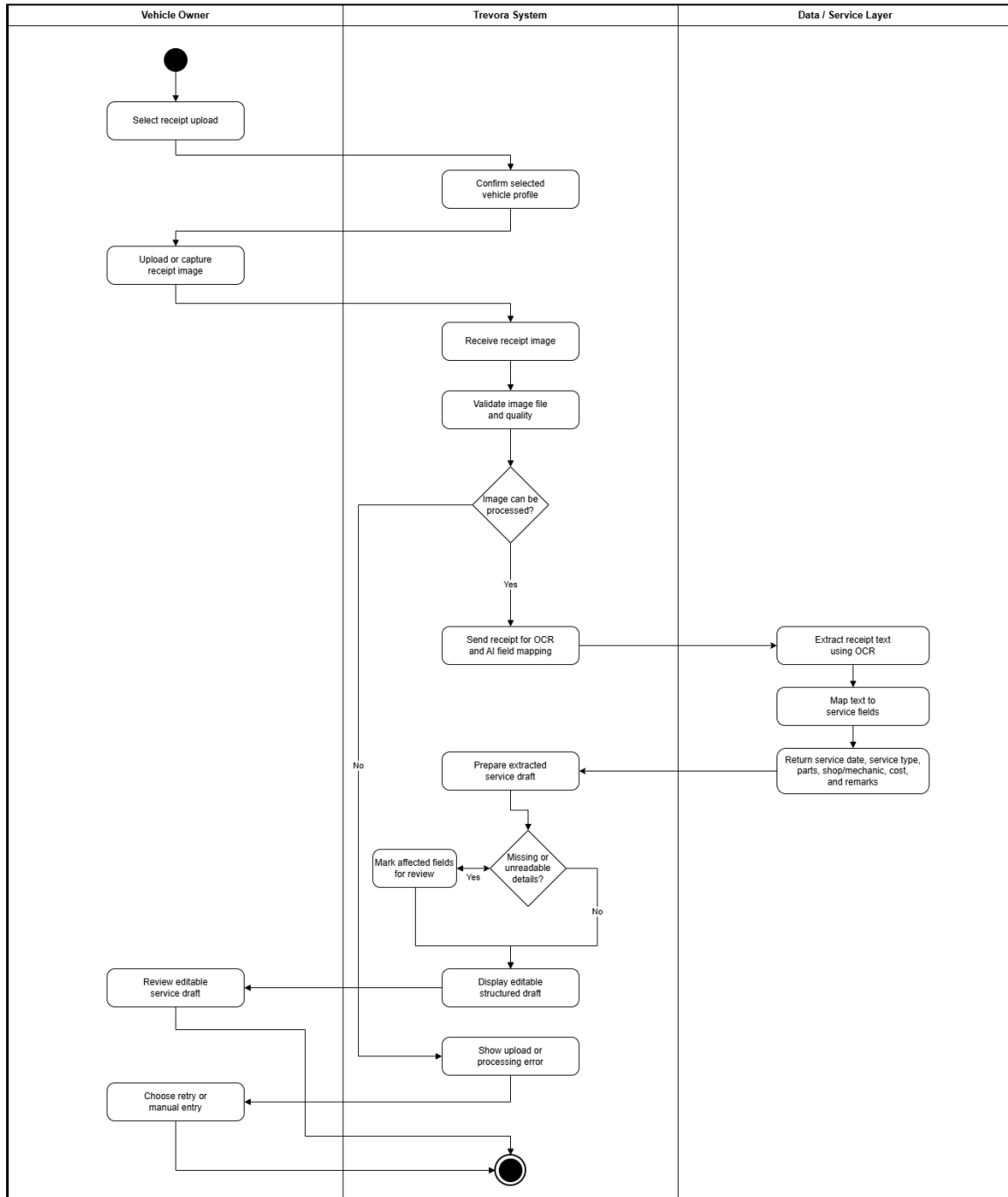


Figure - Activity Diagram for UC-EXT-02 Upload Receipt and Extract Details

Change method
[Receipt]

Add Service Record

OCR is reading text, AI is filling the fields

1 Upload

2 OCR + AI

3 AI Draft

4 Validate

5 Confirm

6 Saved

upload-panel

Upload your receipt

[Supporting copy – file types, max size]

receipt_may07.jpg

✓ Quality: 82% – OCR accuracy will be high
⚡ Quality check fires on upload; warns if image too blurry

[Upload file]

[Use camera]

info-panel

What happens next

1 OCR reads your receipt

2 AI maps text to fields

3 Per-field confidence scored

⚡ Educational panel explains the 3-stage OCR+AI pipeline

Run OCR + AI Analysis →

⚡ CTA disabled until file is uploaded

Figure - Low-Fidelity Wireframe for UC-EXT-02 Upload Receipt and Extract Details

UC-EXT-03 Record Voice Service Information

Use Case Item	Description
Use Case Name	Record Voice Service Information
Related Specific Objective	1.3
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has selected or created a registered vehicle profile.
Trigger	The vehicle owner chooses voice input as the service record input method.
Main Flow	The owner records spoken service information using the device microphone. The system converts the voice input into text using speech-to-text. The system uses AI to formalize the transcript and map available information to service record fields. The system prepares the mapped values as an AI-assisted service draft.
Postcondition	The transcribed and mapped service information is displayed in an editable structured draft for review.
Alternative Flow	If some spoken details are unclear, incomplete, or not mentioned, the system marks the affected fields as uncertain or missing for user correction.

Exception Flow	If the microphone is unavailable, permission is denied, or transcription fails, the system displays an error message and allows the owner to try again or use manual entry.
Requirement ID	FR-EXT-03

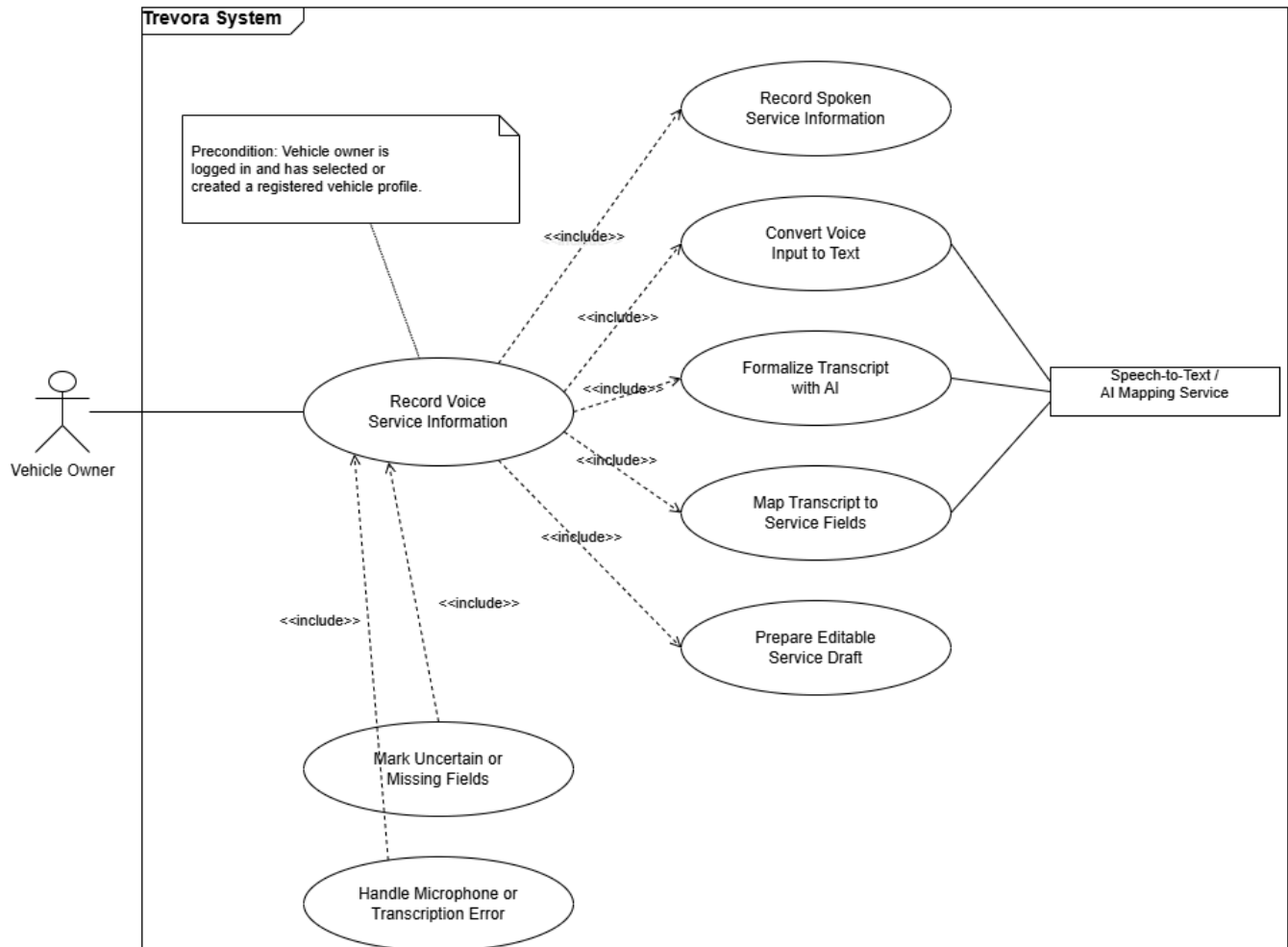


Figure - Use Case Diagram for UC-EXT-03 Record Voice Service Information

Activity Diagram - UC-EXT-03
Record Voice Service Information

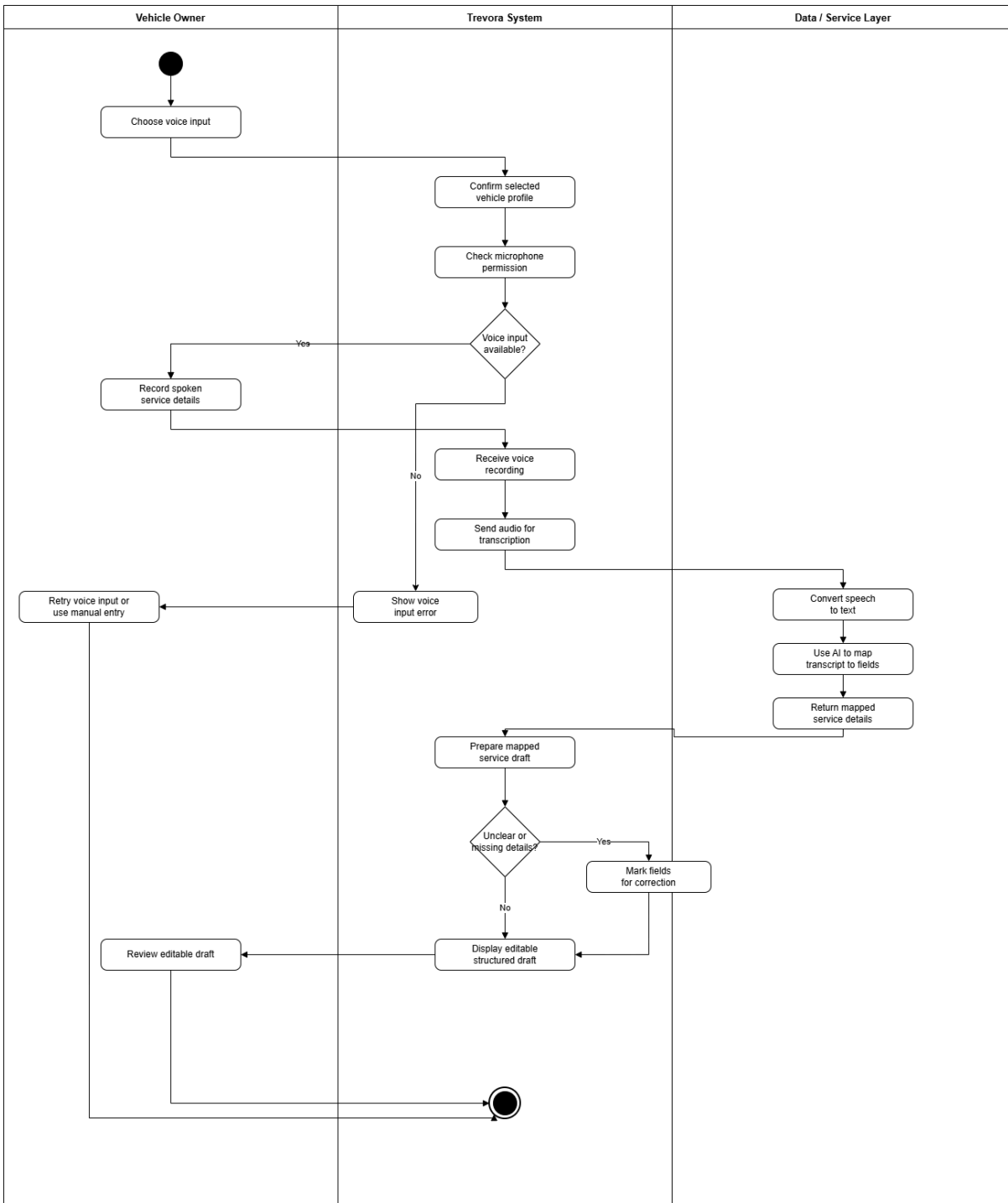


Figure - Activity Diagram for UC-EXT-03 Record Voice Service Information

Change method
[Voice]

Add Service Record

AI is transcribing and formalising your voice note

1
Record

2
AI Process

3
Review

4
Validate

5
Confirm

6
Saved

voice-panel

Describe your service

[Supporting copy – speak naturally]

✓ DONE

Recording complete

[Re-record]

Mic button toggles state; auto-stops with mock transcript after ~4s

TRANSCRIPT READY

Transcript panel appears once recording stops

expectations

What AI will extract

Service typeHigh accuracy
Shop nameHigh – if mentioned
DateMedium – interpreted
CostLow – verbal estimate
Parts/detailsOften empty
Receipt No.N/A – voice only
↳ Sets expectations before AI processes – reduces frustration at review

[⚠ Voice records are less detailed]

Formalise with AI →

↳ Disabled until transcript is available

Figure - Low-Fidelity Wireframe for UC-EXT-03 Record Voice Service Information

UC-EXT-04 Enter Service Details Manually

Use Case Item	Description
Use Case Name	Enter Service Details Manually
Related Specific Objective	1.4
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has selected or created a registered vehicle profile.
Trigger	The owner chooses manual entry, or receipt image upload/voice input is unavailable.
Main Flow	The owner manually enters service details such as service date, service type, total cost, odometer, shop/mechanic name, location, parts replaced, labor/work performed, and remarks when available. The system checks whether the required fields are filled before allowing the owner to continue.
Postcondition	A structured owner-entered service draft is prepared for confirmation.
Alternative Flow	If optional details are unavailable, the owner may leave those fields blank and continue as long as all required fields are complete.

Exception Flow	If a required field is missing, the system highlights the missing field and prompts the owner to complete it before continuing.
Requirement ID	FR-EXT-04

Use Case Diagram - UC-EXT-04

Enter Service Details Manually

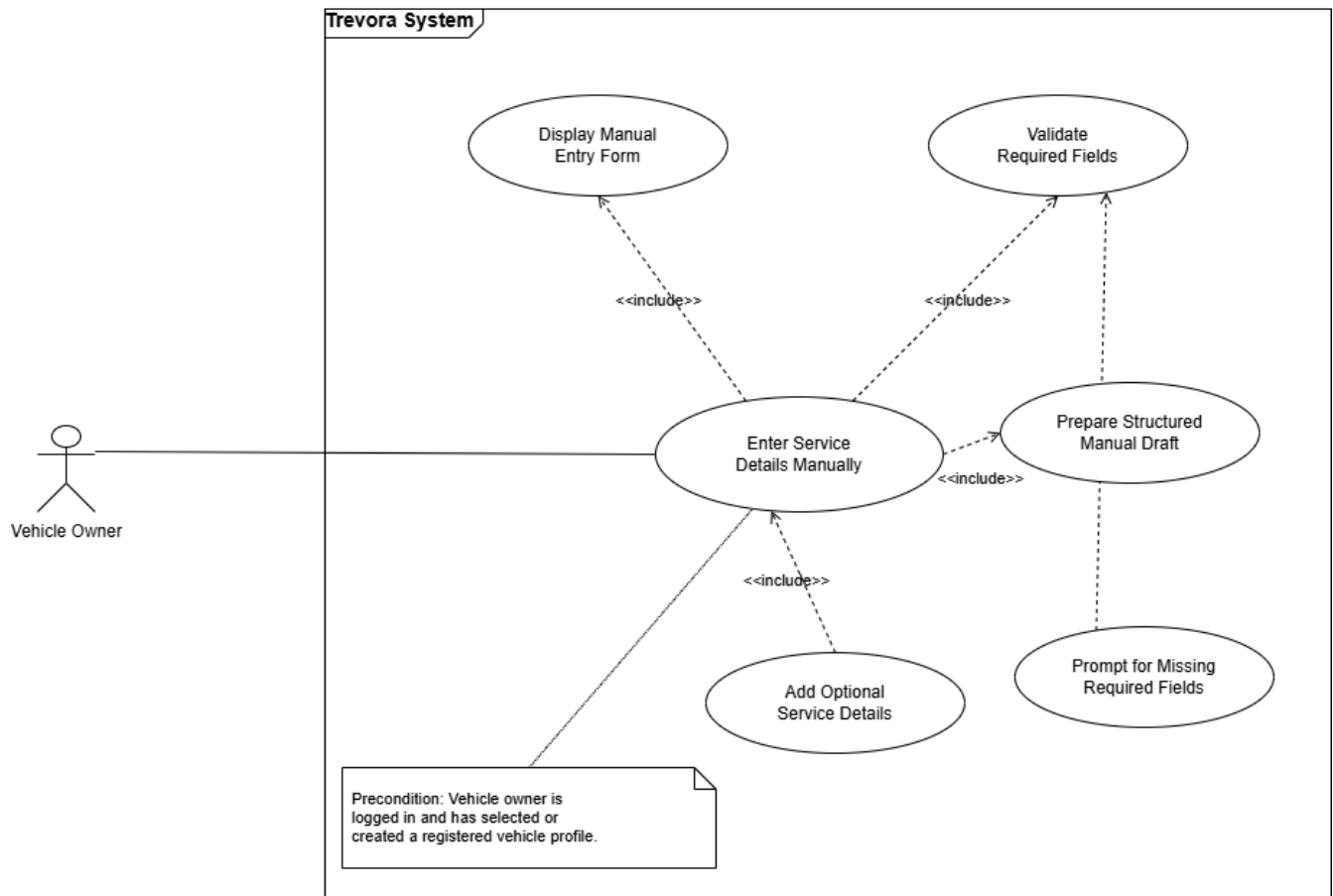


Figure - Use Case Diagram for UC-EXT-04 Enter Service Details Manually

Activity Diagram - UC-EXT-04

Enter Service Details Manually

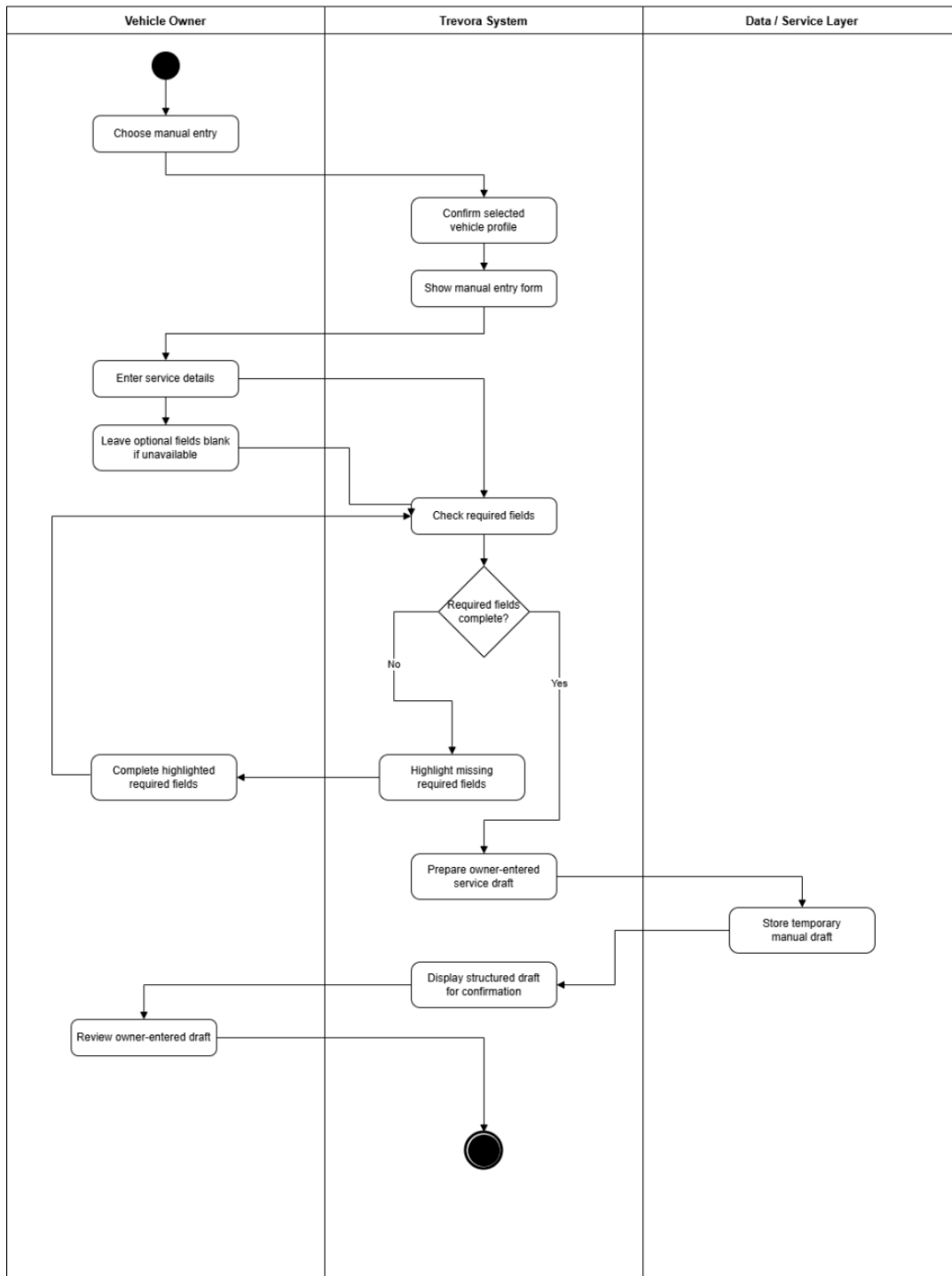


Figure - Activity Diagram for UC-EXT-04 Enter Service Details Manually

1

Fill Details

2

Confirm

3

Saved

manual-form

Enter service details

Owner-verified

Vehicle *

Service Date *

Service Type *

Total Cost *

Odometer (km)

Shop / Mechanic

Shop Location

Parts Replaced

Labor / Work Performed

Remarks

owner-verified-info

Owner-verified entries

Trust signal – differentiates manual from AI-assisted entries

[No AI processing]

required-status

Required fields

Vehicle

Service Date

Service Type

Total Cost

Required

Required

Required

Required

Live checklist – gates the CTA

Review & Confirm →

2-col grid: half-width for short fields, full-width for descriptive

Figure - Low-Fidelity Wireframe for UC-EXT-04 Enter Service Details Manually

UC-EXT-05 Convert Input to Structured Service Entry

Use Case Item	Description
Use Case Name	Convert Input to Structured Service Entry
Related Specific Objective	1.5
Primary Actor	Vehicle Owner
Precondition	The vehicle owner has selected or created a registered vehicle profile and has submitted service information through receipt image upload, voice input, or manual entry.
Trigger	The system receives service information from any input method.
Main Flow	The system converts the submitted information into one structured service entry format containing the required service record fields, including vehicle profile, service date, service type, cost, and other relevant service details. For receipt and voice input, the system includes field source and confidence information where applicable. For manual input, the system marks the entry as owner-entered.
Postcondition	A structured service draft is created and displayed for review or confirmation.

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Alternative Flow	If some values are incomplete, the system keeps the fields blank and marks them for validation or required-field completion.
Exception Flow	If the submitted input cannot be processed, the system displays an error message and allows the owner to retry or use manual entry.
Requirement ID	FR-EXT-05

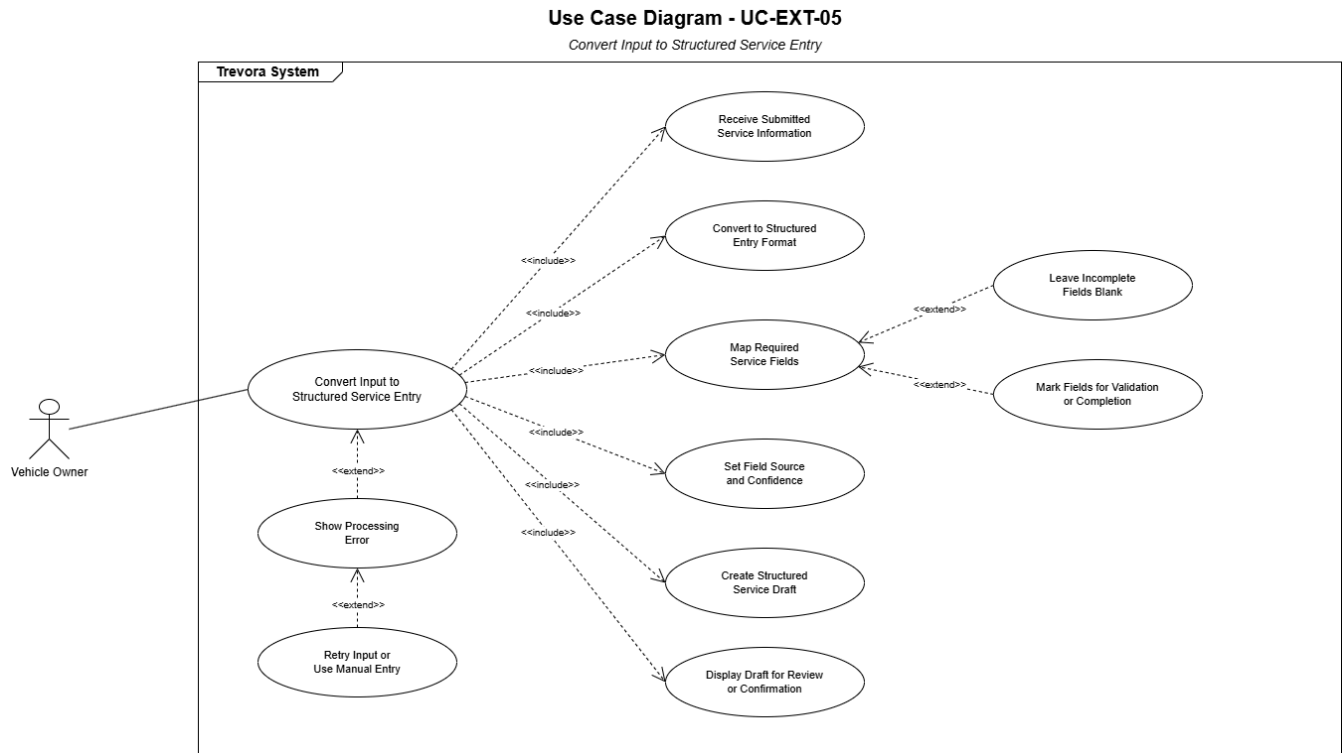


Figure - Use Case Diagram for UC-EXT-05 Convert Input to Structured Service Entry

Activity Diagram - UC-EXT-05
Convert Input to Structured Service Entry

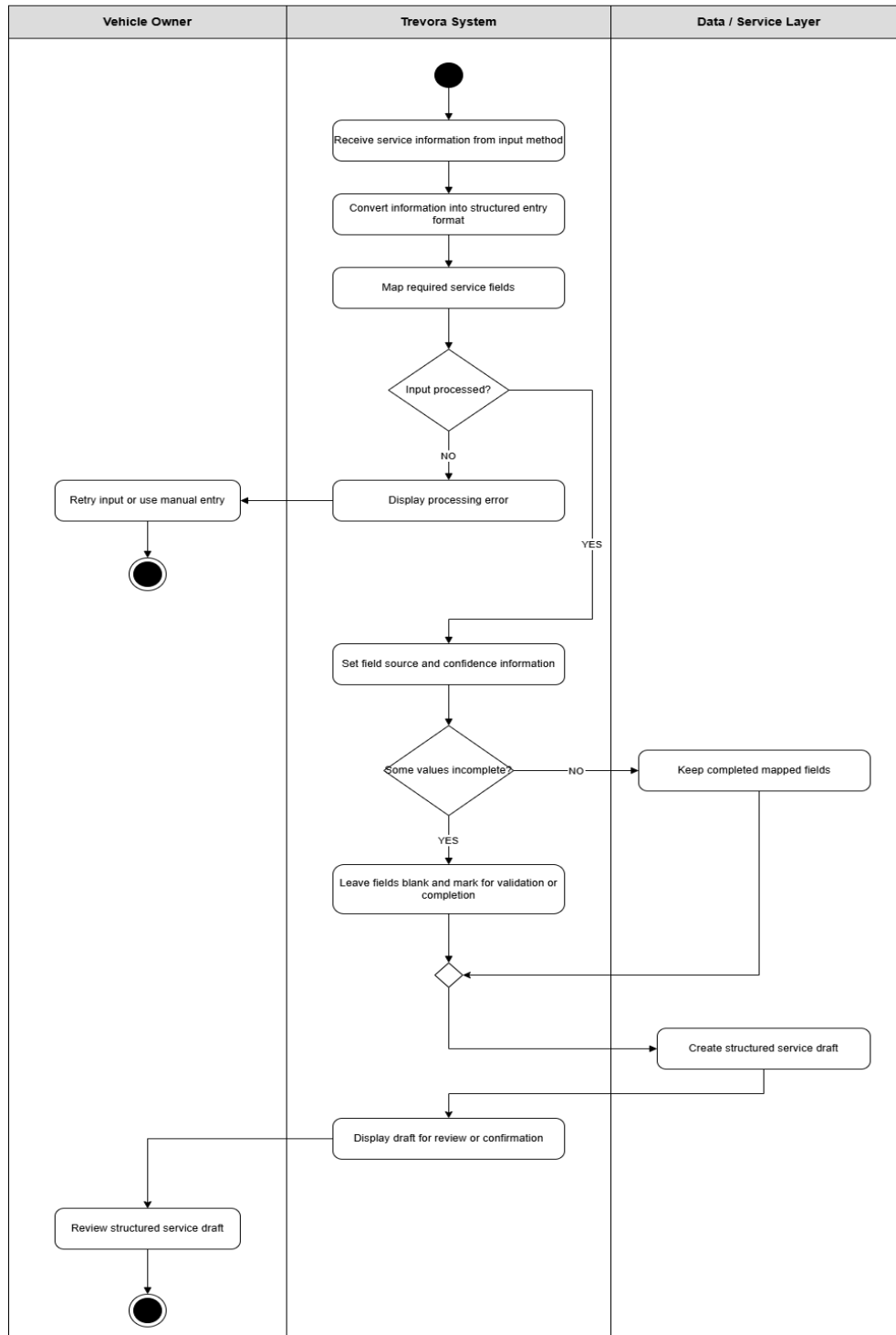


Figure - Activity Diagram for UC-EXT-05 Convert Input to Structured Service Entry

Change method

[Receipt]

Add Service Record

Correct low-confidence and missing fields before saving

1

2

3

4

5

6

Upload

OCR + AI

AI Draft

Validate

Confirm

Saved

ai-summary

[AI extraction complete]

✓ High: 4

~ Medium: 3

! Low: 2

? Not found: 2

↳ Confidence distribution bar – drives CTA Label below

Receipt preview

Hover → highlight

SUPERIOR AUTO REPAIRS

123 Kattipung, Bua, OC

09 May 2026 08:45:00

Date: 07 May 2026

SERVICES REQUIRED

1. Engine Oil Change

2. Brake Pad Replacement (Fr + Br)

Oil Filter 1x P100

Brake Pads (Fr) 1x P1,200

Brake Pads (Br) 1x P1,200

LABOR P1,000

TOTAL P7,850

Auto-filled fields

Vehicle *

✓ High

Toyota Vios 2021

↳ Pre-selected

Service Date *

✓ High

05/07/2026

↳ Line 2: '07 May 2026'

Service Type *

✓ High

Oil Change; Brake Pad Replacement

↳ Lines 5-6

Total Cost *

✓ High

PHP 7,850.00

↳ Total line

Figure - Low-Fidelity Wireframe for UC-EXT-05 Convert Input to Structured Service Entry

Module 2: Service Data Validation and Correction

General Objective 2: Improve the reliability of saved service records by requiring user review, validation, correction, and confirmation before saving, reducing incorrect or incomplete saved entries by at least 30%.

Specific Objective-to-Requirement Mapping

Specific Objective	Use Case / Function	Requirement ID
2.1	UC-VAL-01 Review Service Draft	FR-VAL-01
2.2	UC-VAL-02 Identify Missing Required and Flagged Fields	FR-VAL-02
2.3	UC-VAL-03 Correct Flagged or Incomplete Service Details	FR-VAL-03
2.4	UC-VAL-04 Confirm and Save Validated Record	FR-VAL-04

Requirement ID	Requirement Statement
FR-VAL-01	The system shall display extracted, AI-assisted, or owner-entered service details in an editable review form before saving.
FR-VAL-02	The system shall identify missing required fields such as vehicle profile, service date, service type, and cost before submission, and shall flag low-confidence or not-found fields for receipt-based and voice-based service drafts.
FR-VAL-03	The system shall allow users to correct inaccurate, incomplete, missing, low-confidence, or not-found service details before saving.
FR-VAL-04	The system shall require user confirmation before storing validated service records under the selected registered vehicle profile.

UC-VAL-01 Review Service Draft

Use Case Item	Description
Use Case Name	Review Service Draft
Related Specific Objective	2.1
Primary Actor	Vehicle Owner
Precondition	A structured service draft has been created from receipt image upload, voice input, or manual entry.
Trigger	The vehicle owner opens the service draft for review.
Main Flow	The system displays the service draft in an editable review form. For receipt-based and voice-based drafts, the system shows extracted or AI-assisted field values with their source and confidence status when available. For manual drafts, the system displays owner-entered values without AI confidence scoring. The owner reviews the service details before proceeding.
Postcondition	The service draft is ready for required-field checking, confidence-field checking, correction, or confirmation.
Alternative Flow	If the owner finds incorrect details, the owner edits the fields before proceeding.
Exception Flow	If the service draft cannot be loaded, the system displays an error message and allows the owner to retry or return to service record input.
Requirement ID	FR-VAL-01

Use Case Diagram - UC-VAL-01

Service Data Validation and Correction

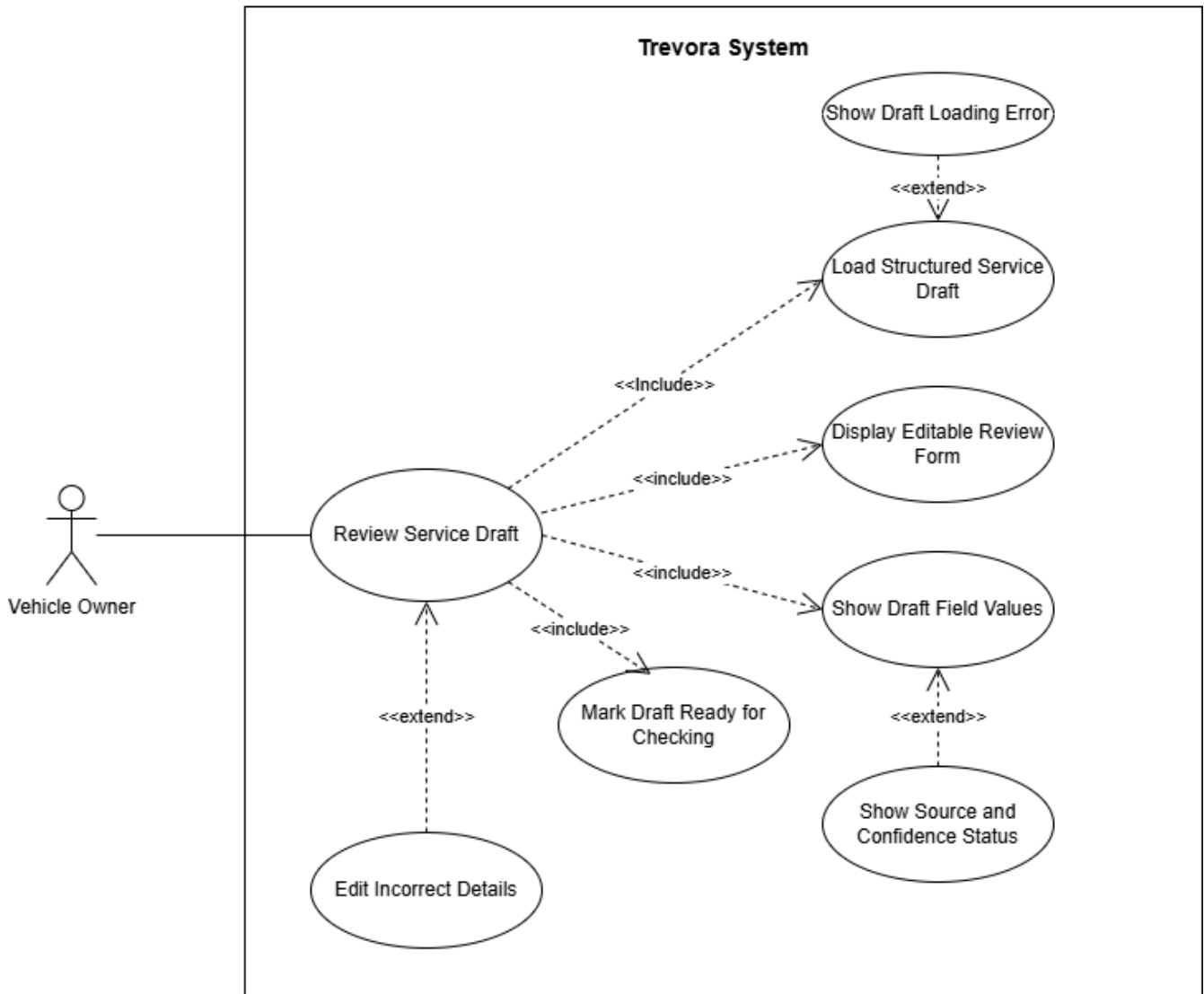


Figure - Use Case Diagram for UC-VAL-01 Review Service Draft

Activity Diagram - UC-VAL-01
Review Service Draft

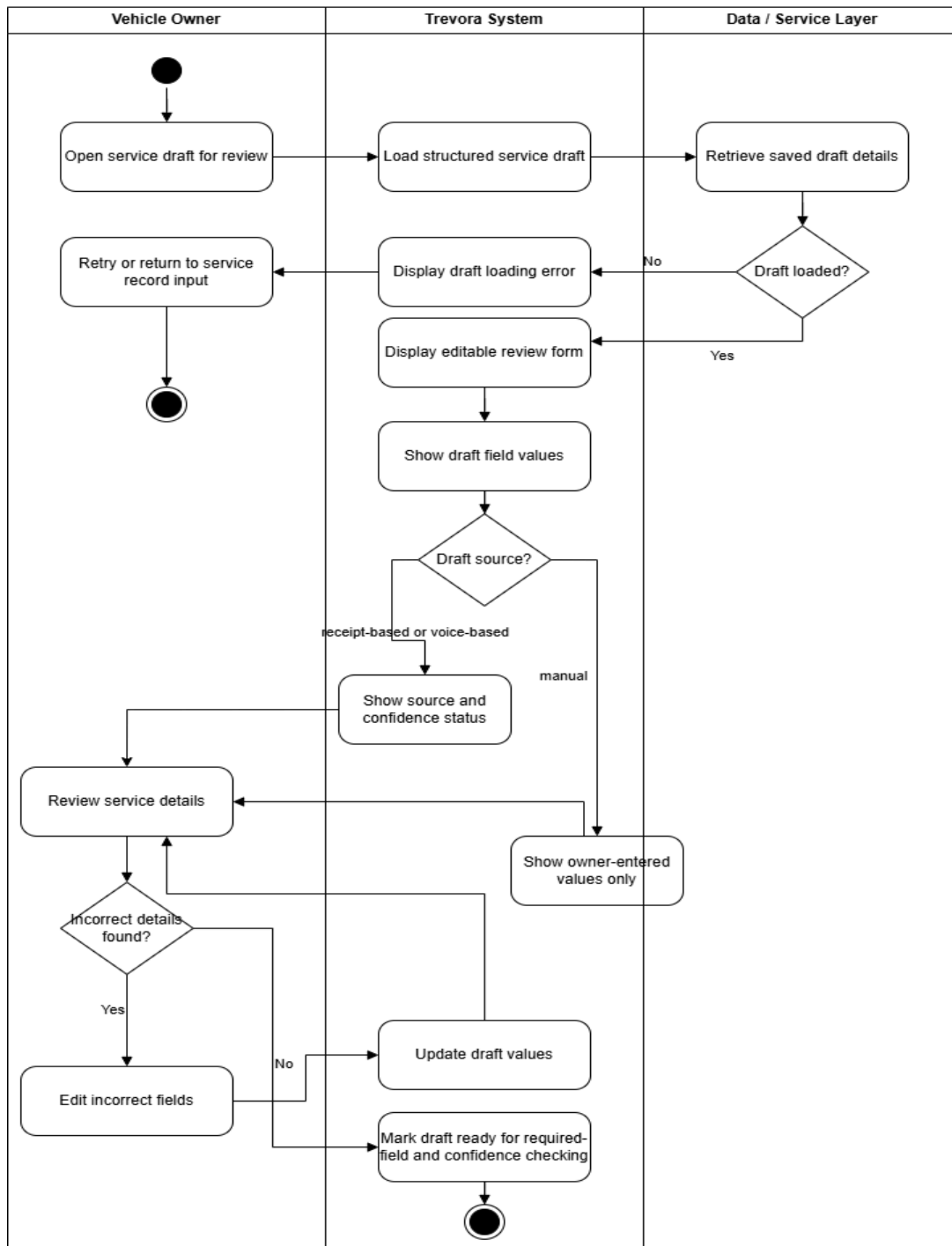


Figure - Activity Diagram for UC-VAL-01 Review Service Draft

← Change method

[Receipt]

Add Service Record

Correct low-confidence and missing fields before saving

1

Upload

2

OCR + AI

3

AI Draft

4

Validate

5

Confirm

6

Saved

ai-summary

[AI extraction complete]

✓ High: 4

~ Medium: 3

! Low: 2

? Not found: 2

↳ Confidence distribution bar – drives CTA Label below

Receipt preview

Hover → highlight

SUPERIOR AUTO REPAIRS

123 Kaitiaki Ave, GC

09 123 456 7890

Date: 07 May 2026

SERVICE REQUESTED

1. Engine oil change

2. Brake Pad Replacement (Fr + Rr)

Oil Filter & ... PHP0

Brake Pads (Fr) ... PHP200

Brake Pads (Rr) ... PHP200

Labor ... PHP100

TOTAL ... PHP500

Auto-filled fields

Vehicle *

✓ High

Toyota Vios 2021

↳ Pre-selected

Service Date *

✓ High

05/07/2026

↳ Line 2: '07 May 2026'

Service Type *

✓ High

Oil Change; Brake Pad Replacement

↳ Lines 5-6

Total Cost *

✓ High

PHP 7,850.00

↳ Total line

Figure - Low-Fidelity Wireframe for UC-VAL-01 Review Service Draft

UC-VAL-02 Identify Missing Required and Flagged Fields

Use Case Item	Description
Use Case Name	Identify Missing Required and Flagged Fields
Related Specific Objective	2.2
Primary Actor	Vehicle Owner
Precondition	A structured service draft exists and contains missing, inaccurate, incomplete, low-confidence, or not-found details.
Trigger	The owner attempts to proceed from review, validation, or confirmation.
Main Flow	The system checks whether required fields such as vehicle profile, service date, service type, and cost are complete before saving. For receipt-based and voice-based drafts, the system also checks field confidence levels and flags low-confidence or

	not-found fields for user validation. For manual drafts, the system checks required fields only and treats completed fields as owner-entered.
Postcondition	Missing required fields and low-confidence or not-found fields are highlighted for review or correction.
Alternative Flow	If no missing required fields or flagged fields are found, the system allows the owner to proceed to confirmation.
Exception Flow	If the field check cannot be completed, the system displays an error message and keeps the service draft unsaved.
Requirement ID	FR-VAL-02

Use Case Diagram - UC-VAL-02

Service Data Validation and Correction

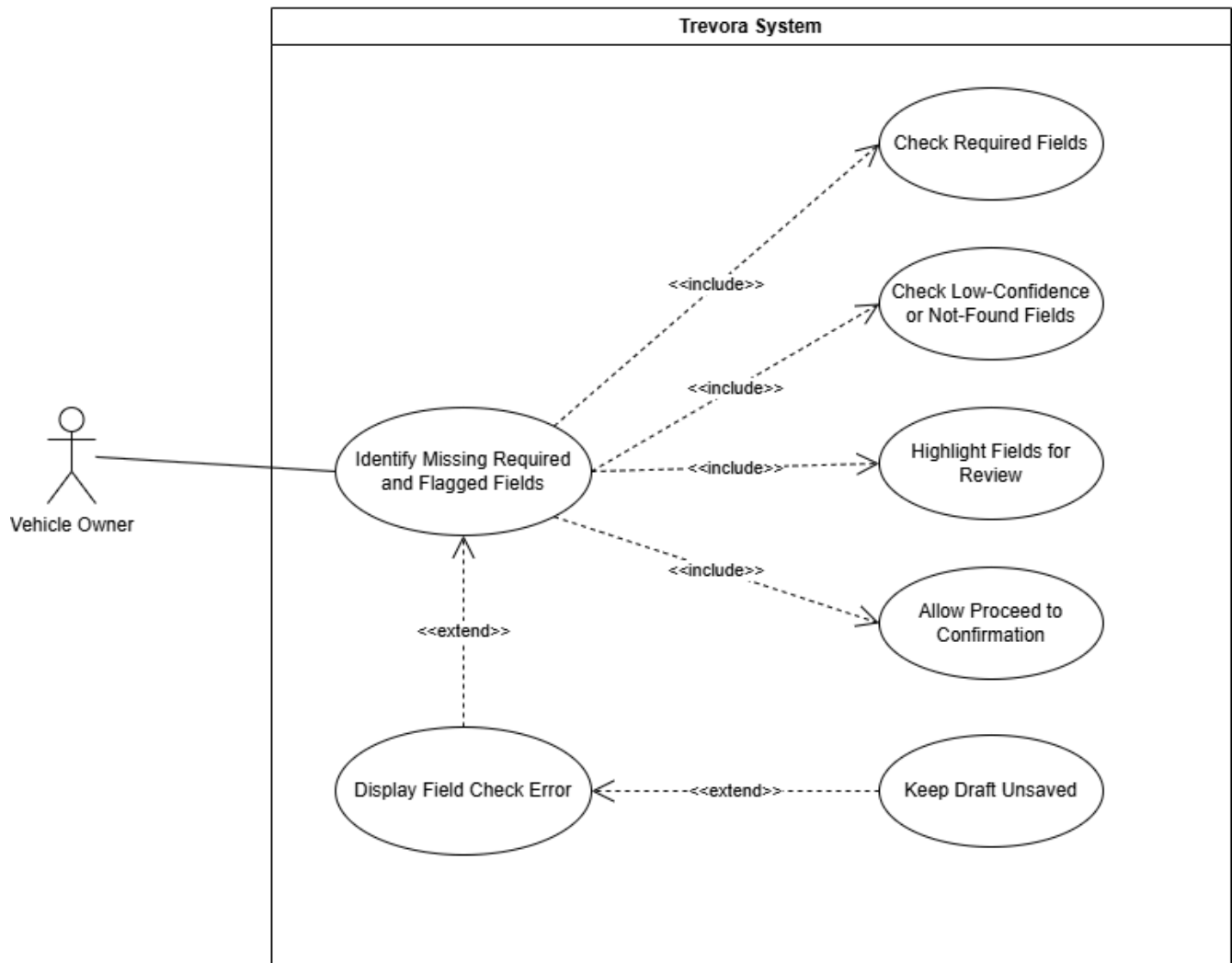


Figure - Use Case Diagram for UC-VAL-02 Identify Missing Required and Flagged Fields

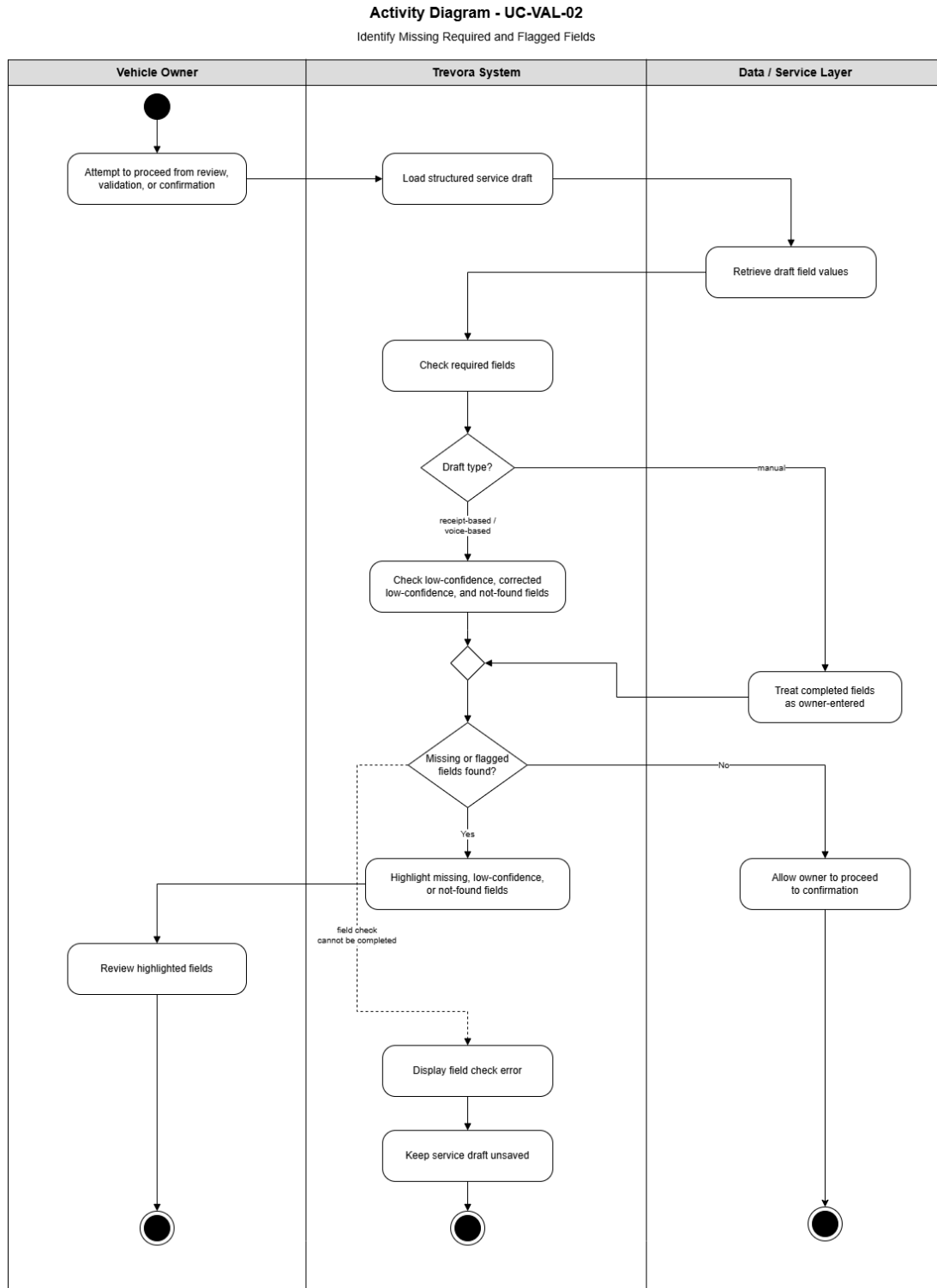


Figure - Activity Diagram for UC-VAL-02 Identify Missing Required and Flagged Fields

← Change method [Receipt]

Add Service Record

Confirm the validated record before adding it to history

1Upload

2OCR + AI

3AI Draft

4Validate

5Confirm

6Saved

[⚠ 4 fields need attention]

⚡ Warning banner – only shown when flagged fields exist

field-status

Field status

Vehicle

✓

Service Date

✓

Service Type

✓

Total Cost

✓

Shop / Mechanic

✓

Parts Replaced

✓

Labor / Work Performed

✓

Shop Location

✓

Odometer (km)

? Not found

Receipt No.

✓

Remarks

? Not found

⚡ Persistent sidebar overview of all field states

Proceed to Confirm →

Correct these fields

Labor / Work Performed

! Low

Drain & refill, pad install

⚡ AI inferred

Shop Location

! Low

Katipunan Ave, QC

⚡ Low-contrast footer

Odometer (km)

? Not found

[Not found – enter manually]

⚡ Not detected

Remarks

? Not found

[Not found – enter manually]

⚡ Not detected

Figure - Low-Fidelity Wireframe for UC-VAL-02 Identify Missing Required and Flagged Fields

UC-VAL-03 Correct Flagged or Incomplete Service Details

Use Case Item	Description
Use Case Name	Correct Flagged or Incomplete Service Details
Related Specific Objective	2.3
Primary Actor	Vehicle Owner
Precondition	A structured service draft exists and contains missing, inaccurate, incomplete, low-confidence, or not-found details.
Trigger	The owner chooses to edit the service draft, or the system highlights a field that needs correction.
Main Flow	The owner edits the missing, inaccurate, incomplete, low-confidence, or not-found service details in the review or validation form. The system rechecks the corrected

	fields. For receipt-based and voice-based drafts, corrected low-confidence or not-found fields are updated as user-reviewed or owner-confirmed values.
Postcondition	The corrected service entry is rechecked and prepared for confirmation.
Alternative Flow	If all corrected details are valid and required fields are complete, the system allows the owner to proceed to confirmation.
Exception Flow	If a corrected detail remains invalid or incomplete, the system keeps the field highlighted and prompts the owner to correct it again.
Requirement ID	FR-VAL-03

Use Case Diagram - UC-VAL-03

Service Data Validation and Correction

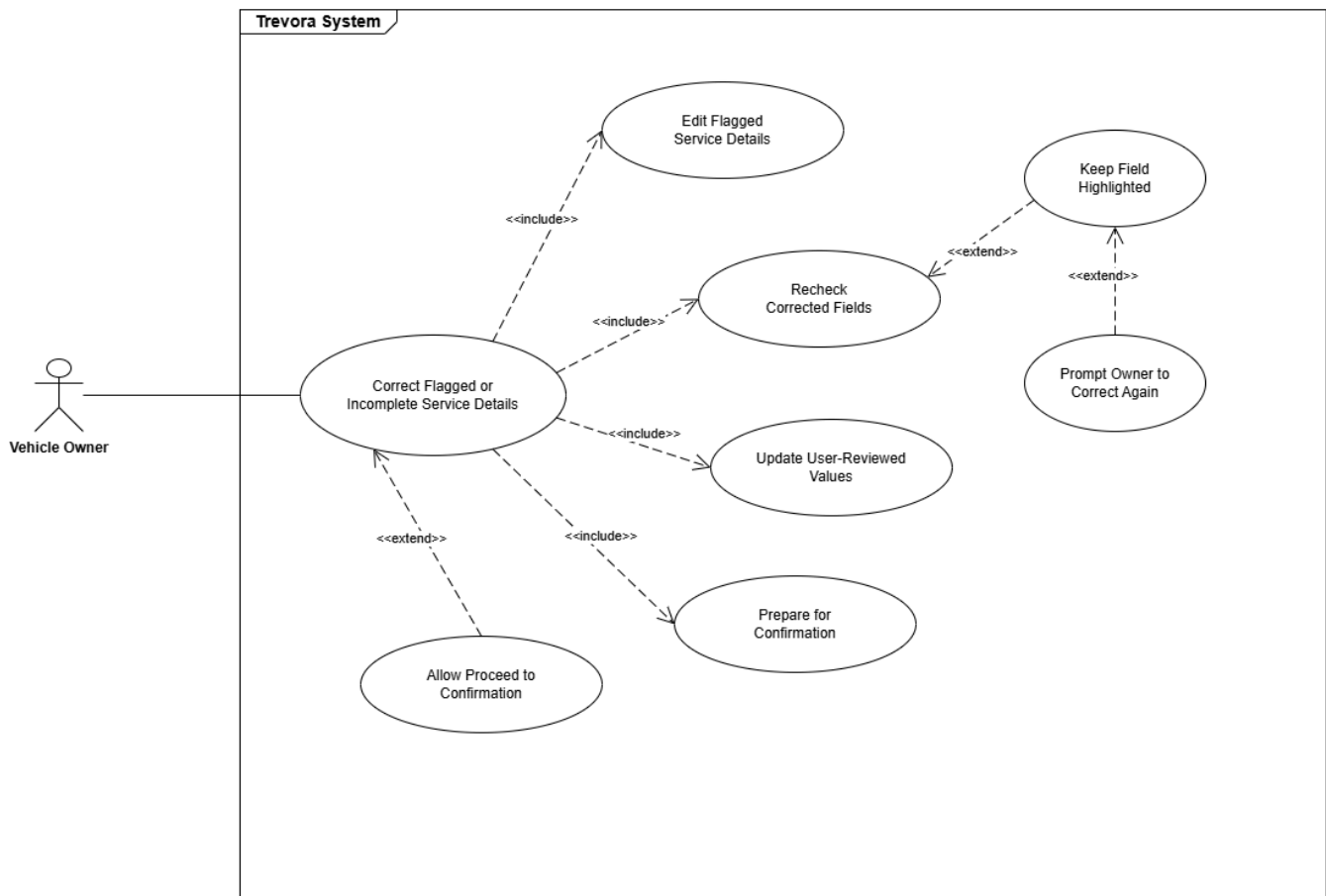


Figure - Use Case Diagram for UC-VAL-03 Correct Flagged or Incomplete Service Details

Activity Diagram - UC-VAL-03

Correct Flagged or Incomplete Service Details

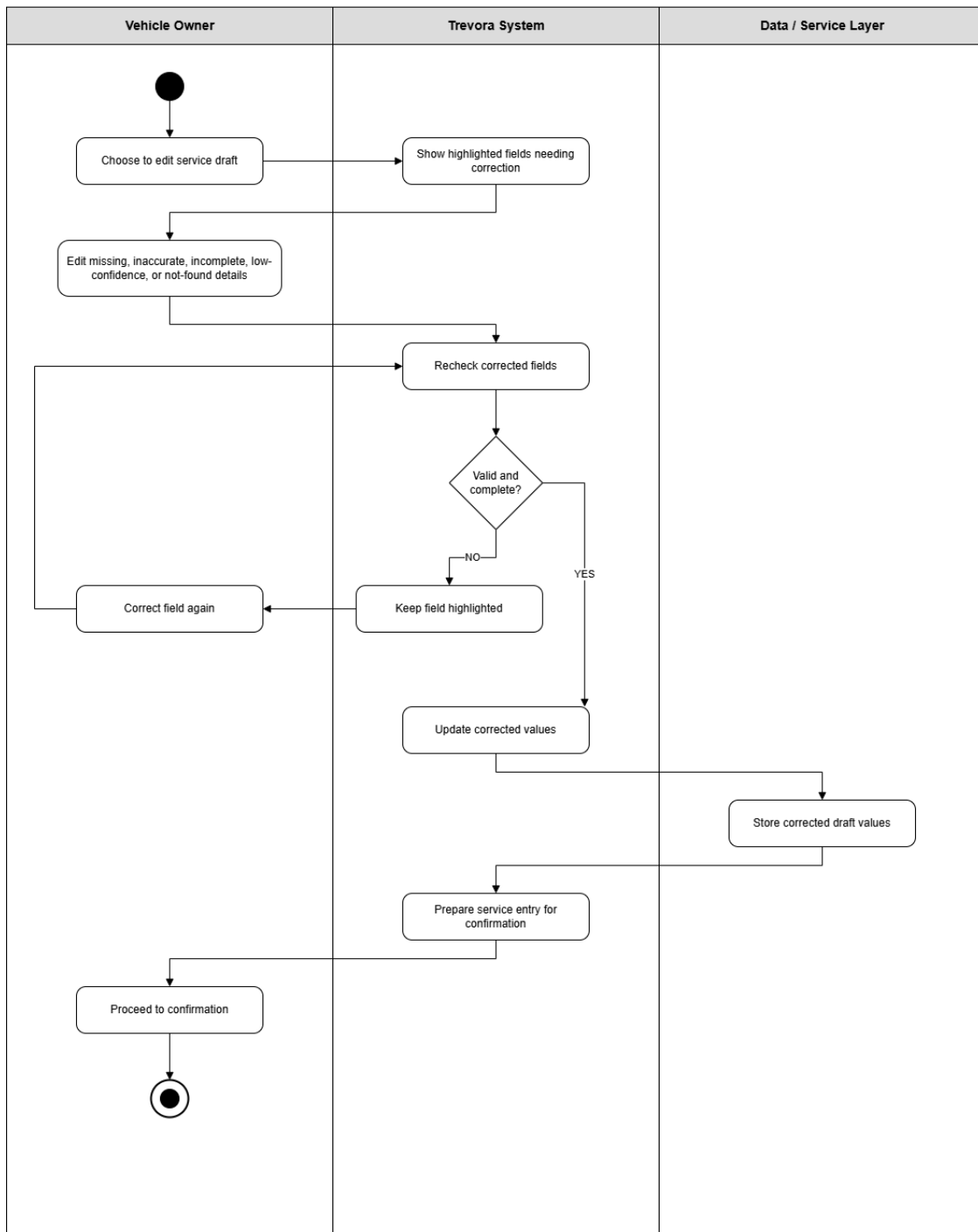


Figure - Activity Diagram for UC-VAL-03 Correct Flagged or Incomplete Service Details

Change method
[Receipt]

Add Service Record

Confirm the validated record before adding it to history

✓

Upload

✓

OCR + AI

✓

AI Draft

4

Validate

5

Confirm

6

Saved

[⚠ 4 fields need attention]

⚡ Warning banner – only shown when flagged fields exist

field-status

Field status

Vehicle	✓
Service Date	✓
Service Type	✓
Total Cost	✓
Shop / Mechanic	✓
Parts Replaced	✓
Labor / Work Performed	✓
Shop Location	✓
Odometer (km)	? Not found
Receipt No.	✓
Remarks	? Not found

⚡ Persistent sidebar overview of all field states

Proceed to Confirm →

Correct these fields

Labor / Work Performed

! Low

Drain & refill, pad install

⚡ AI inferred

Shop Location

! Low

Katipunan Ave, QC

⚡ Low-contrast footer

Odometer (km)

? Not found

[Not found – enter manually]

⚡ Not detected

Remarks

? Not found

[Not found – enter manually]

⚡ Not detected

Figure - Low-Fidelity Wireframe for UC-VAL-03 Correct Flagged or Incomplete Service Details

UC-VAL-04 Confirm and Save Validated Record

Use Case Item	Description
Use Case Name	Confirm and Save Validated Record
Related Specific Objective	2.4
Primary Actor	Vehicle Owner
Precondition	The service draft has complete required fields and has passed validation or required-field checking.
Trigger	The owner chooses to confirm and save the validated service record.
Main Flow	The system displays a final read-only summary of the service record details, including vehicle profile, service date, service type, cost, shop/mechanic name, parts replaced, odometer, and remarks when available. The system shows the record source, such as OCR + AI, Voice + AI, or Owner-entered. The owner reviews the final details and

	confirms that the information is correct. After confirmation, the system stores the validated record under the correct registered vehicle profile.
Postcondition	The validated service record is saved under the correct registered vehicle profile.
Alternative Flow	If the owner cancels or goes back to edit, the record remains unsaved as a draft.
Exception Flow	If saving fails, the system displays an error message and keeps the validated draft available for retry.
Requirement ID	FR-VAL-04

Use Case Diagram - UC-VAL-04

Service Data Validation and Correction

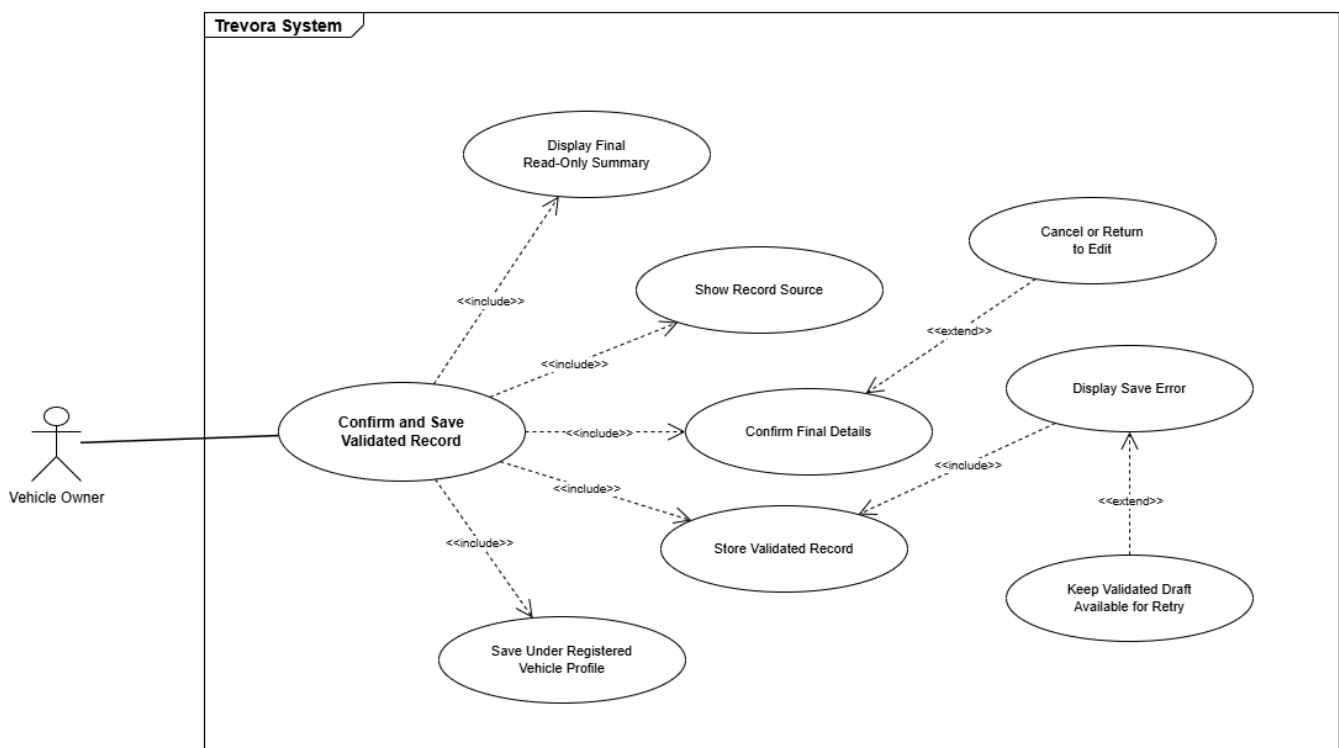


Figure - Use Case Diagram for UC-VAL-04 Confirm and Save Validated Record

Activity Diagram - UC-VAL-04
Confirm and Save Validated Record

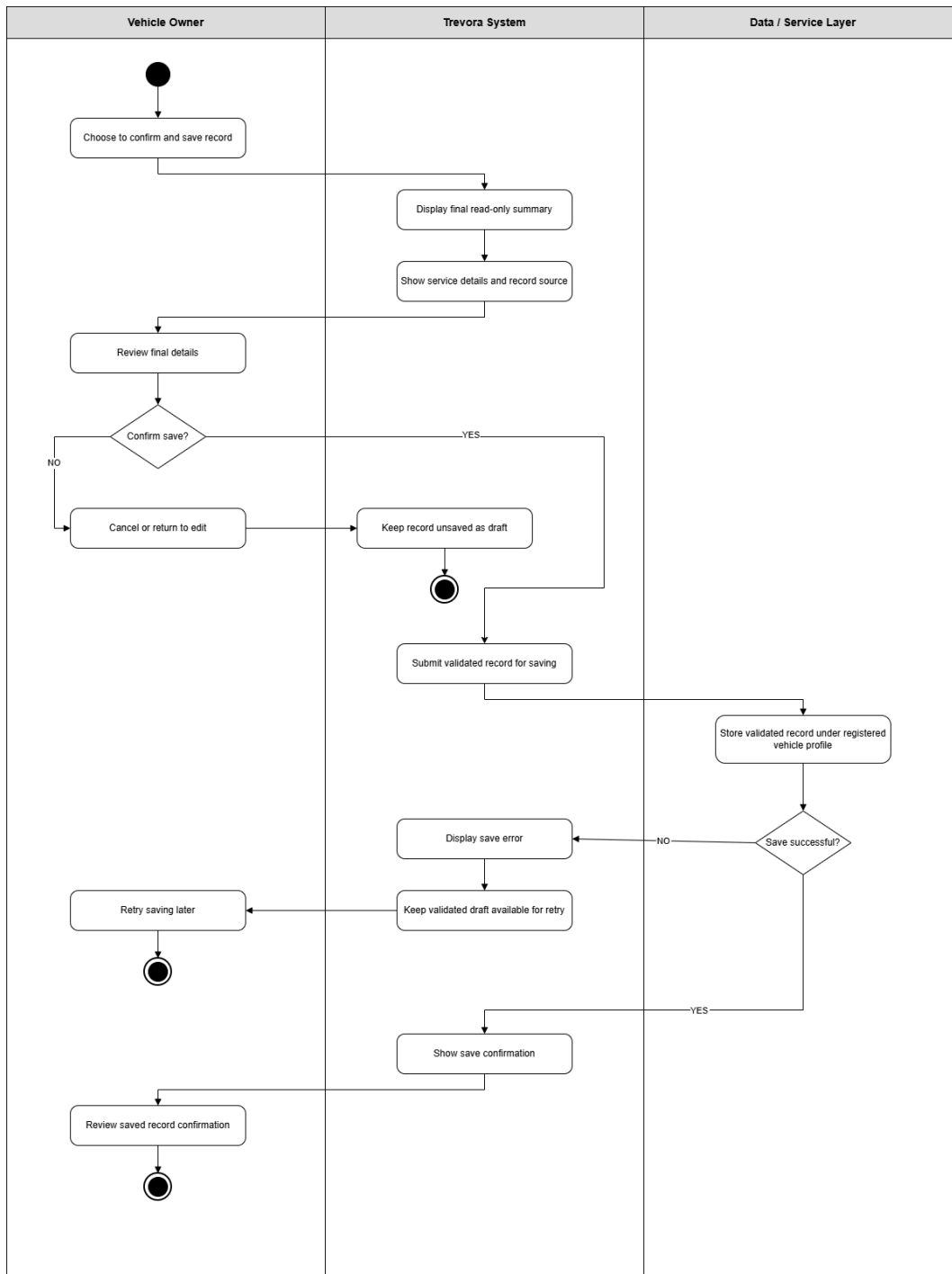


Figure - Activity Diagram for UC-VAL-04 Confirm and Save Validated Record

← Change method [Receipt]

Add Service Record

Your record has been saved to service history

✓ Upload

✓ OCR + AI

✓ AI Draft

✓ Validate

5 Confirm

6 Saved

Final record summary

[Will be saved to Toyota Vios 2021]

[OCR + AI]

Vehicle	Toyota Vios 2021	✓ High
Service Date	05/07/2026	✓ High
Service Type	Oil Change; Brake Pad Replacement	✓ High
Cost	PHP 7,850.00	✓ High
Shop	Superior Auto Repairs	~ Medium
Parts	Oil filter, brake pads (front)	~ Medium
Odometer	? Not found	
Remarks	? Not found	

↳ Read-only summary table – final review before permanent save

☒ I confirm the details above are correct and authorise this record to be added to my vehicle's permanent service history.

↳ Explicit consent checkbox – required before save button activates

← Back to Edit

[Confirm & Save Record]

Figure - Low-Fidelity Wireframe for UC-VAL-04 Confirm and Save Validated Record

Module 3: Unified Vehicle Service History Consolidation

General Objective 3: Reduce fragmented vehicle service data by consolidating 100% of validated maintenance and repair records under the correct registered vehicle profile in one chronological service history.

Specific Objective-to-Requirement Mapping

Specific Objective	Use Case / Function	Requirement ID
3.1	UC-HIS-01 Link Validated Record to Vehicle Profile	FR-HIS-01
3.2	UC-HIS-02 Organize Records Chronologically	FR-HIS-02
3.3	UC-HIS-03 Categorize Service Records	FR-HIS-03
3.4	UC-HIS-04 View Unified Vehicle Service History	FR-HIS-04

Requirement ID	Requirement Statement
FR-HIS-01	The system shall store all validated service records under the correct registered vehicle profile.
FR-HIS-02	The system shall organize saved records chronologically by service date.
FR-HIS-03	The system shall categorize saved records by service type, parts replaced, cost, and shop/mechanic details.
FR-HIS-04	The system shall display the complete validated maintenance and repair history of each registered vehicle in one centralized record.

UC-HIS-01 Link Validated Record to Vehicle Profile

Use Case Item	Description
Use Case Name	Link Validated Record to Vehicle Profile
Related Specific Objective	3.1
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in, has a registered vehicle profile, and has confirmed a validated service record.
Trigger	A validated record is ready to be saved.
Main Flow	The system stores the validated service record under the selected registered vehicle profile.
Postcondition	The validated service record is placed under the correct vehicle profile.
Alternative Flow	If the selected vehicle profile is unavailable, the system allows the owner to select another registered vehicle profile before saving.
Exception Flow	If the vehicle profile cannot be verified or the record cannot be linked, the system displays an error message and keeps the validated record unsaved for retry.
Requirement ID	FR-HIS-01

Use Case Diagram - UC-HIS-01

Unified Vehicle Service History Consolidation

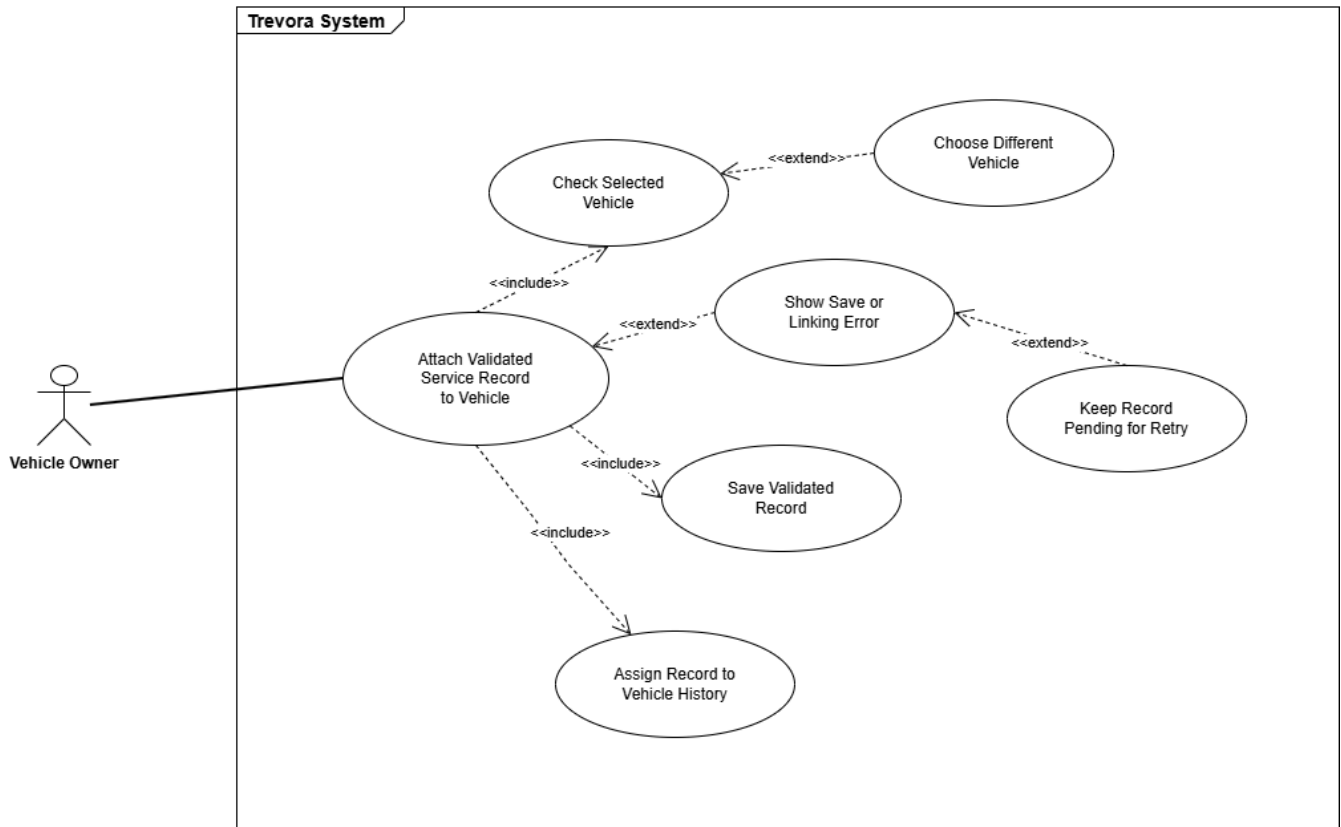


Figure - Use Case Diagram for UC-HIS-01 Link Validated Record to Vehicle Profile

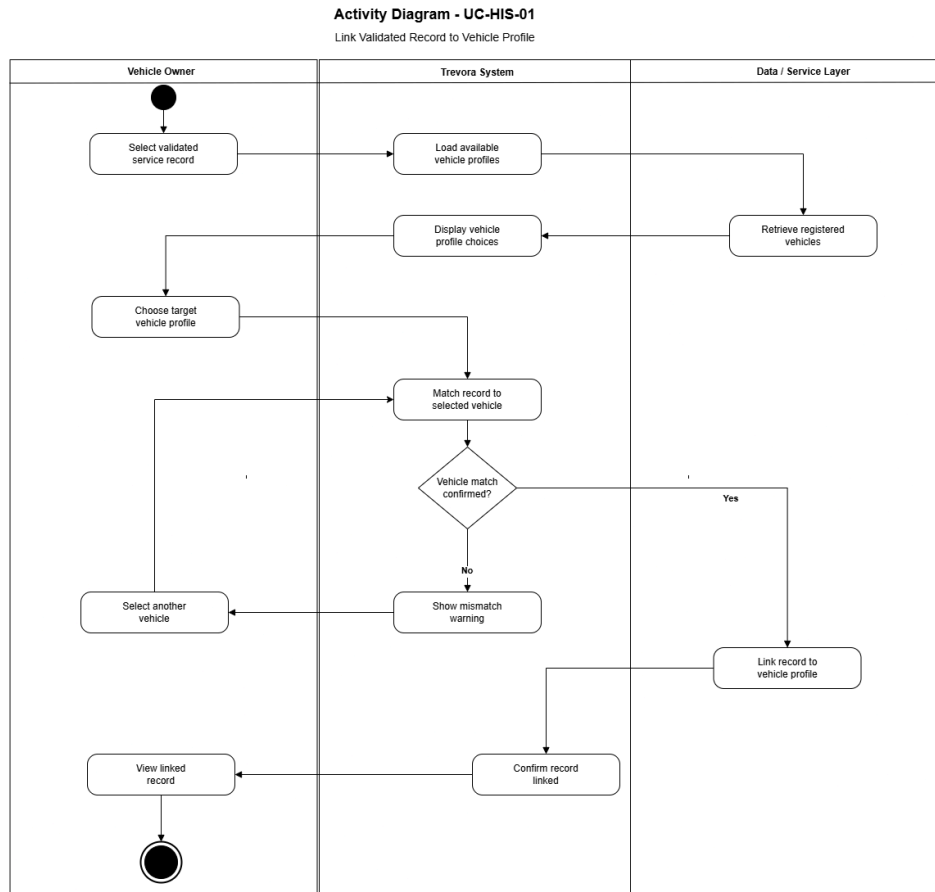


Figure - Activity Diagram for UC-HIS-01 Link Validated Record to Vehicle Profile

WF Service History – Low-fidelity wireframe

Service History

Toyota Vios 2021 • 12 records

[+ Add Record]

Search service type, shop, parts...

[Filters]

[List]

[Timeline]

[Show Empty State]

Toggle buttons switch between List and Timeline views; Filters button shows/hides filter panel

Date	Service Type	Category	Shop	Odometer	Cost	Source	Status
05/07/2026	Oil Change	Maintenance	Superior Auto	62,400 km	₱7,850	Receipt	Verified
03/15/2026	Brake Pad Replacement	Repair	Superior Auto	61,200 km	₱3,200	Manual	Owner
01/02/2026	PMS 40,000 km	Maintenance	Toyota BGC	60,000 km	₱12,500	Receipt	Verified
10/18/2025	Tire Rotation	Maintenance	Giga Tires	58,300 km	₱800	Voice	AI Draft
08/05/2025	Air Filter Replacement	Maintenance	DIY	56,700 km	₱450	Manual	Owner

Clicking a row navigates to Record Detail

Active view: list

Figure - Low-Fidelity Wireframe for UC-HIS-01 Link Validated Record to Vehicle Profile

UC-HIS-02 Organize Records Chronologically

Use Case Item	Description
Use Case Name	Organize Records Chronologically
Related Specific Objective	3.2
Primary Actor	Vehicle Owner
Precondition	The selected registered vehicle has one or more saved service records.
Trigger	The owner opens the vehicle's service history.
Main Flow	The system sorts the saved service records by service date.
Postcondition	The owner sees the vehicle's service records in chronological order.
Alternative Flow	If two or more records have the same service date, the system may sort them by time saved or latest updated record.

Exception Flow	If a service date is missing, the system flags the record for correction before final placement in the timeline.
Requirement ID	FR-HIS-02

Use Case Diagram - UC-HIS-02

Unified Vehicle Service History Consolidation

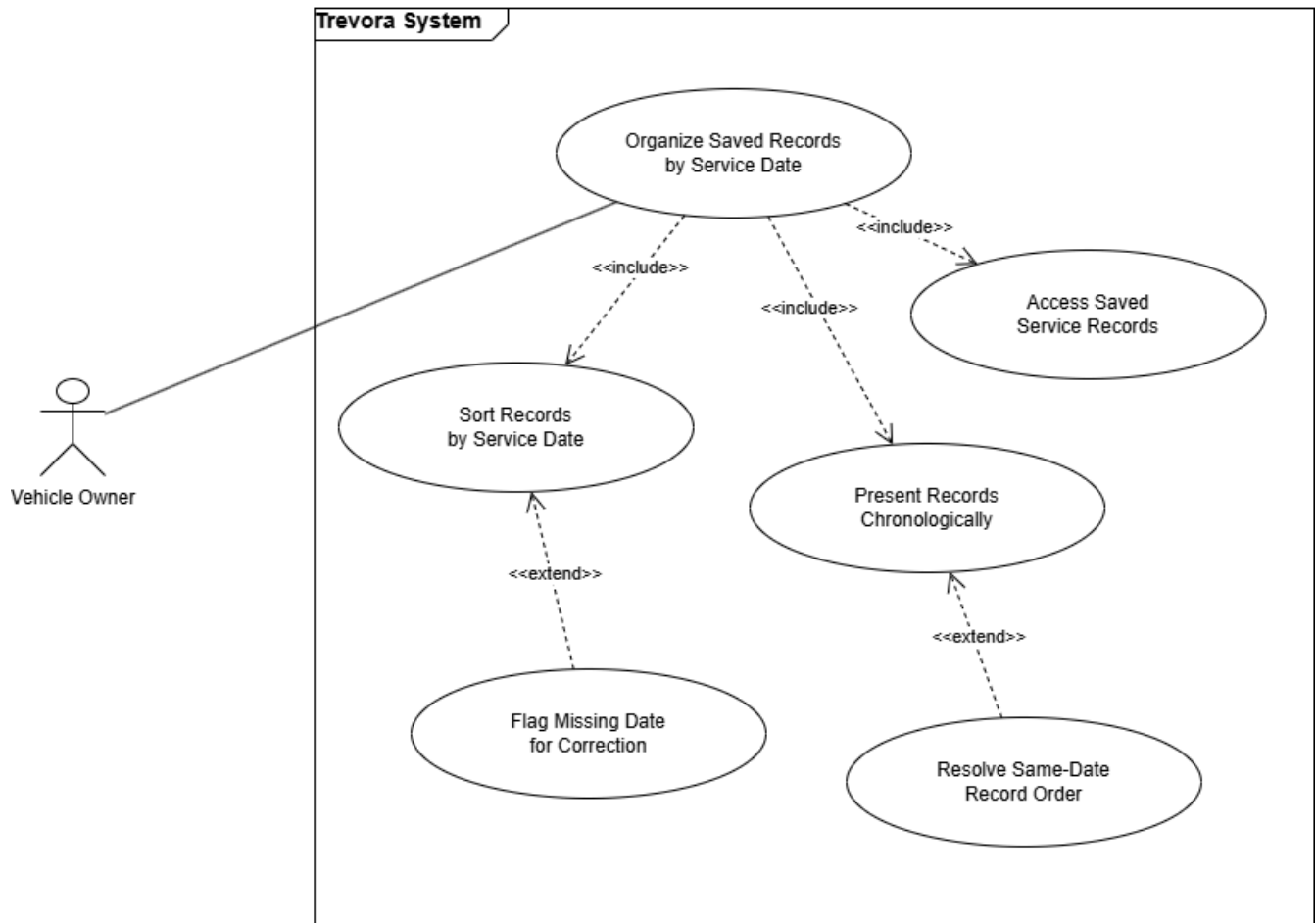


Figure - Use Case Diagram for UC-HIS-02 Organize Records Chronologically

Activity Diagram - UC-HIS-02
Organize Records Chronologically

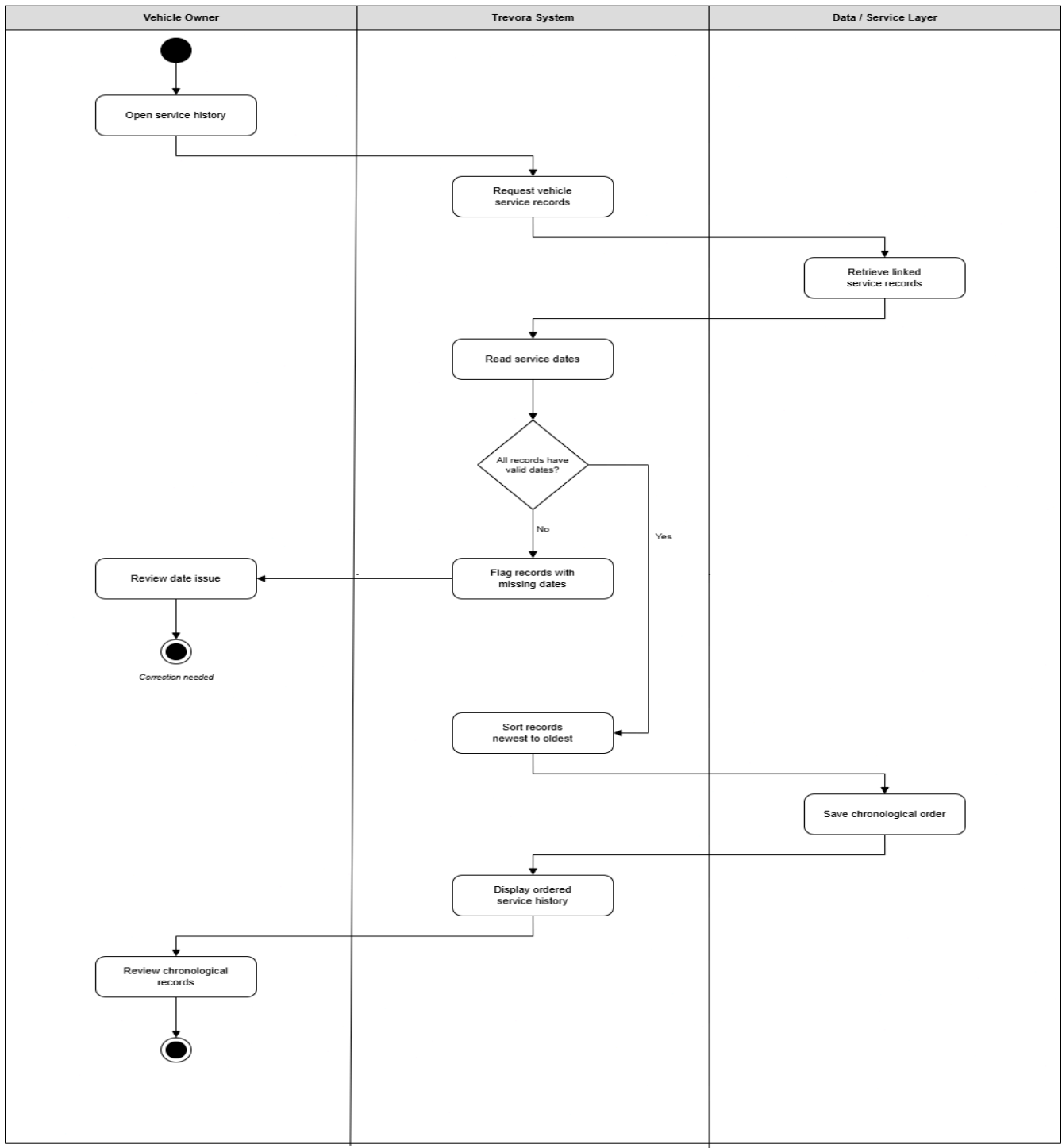


Figure - Activity Diagram for UC-HIS-02 Organize Records Chronologically

Service History

Toyota Vios 2021 · 12 records

[+ Add Record]

[Filters]

[List]

[Timeline]

[Show Empty State]

↳ Toggle buttons switch between List and Timeline views; Filters button shows/hides filter panel

○

○

○

○

↳ Timeline view – vertical chronological list; each card represents one service record

Active view: **timeline**

Figure - Low-Fidelity Wireframe for UC-HIS-02 Organize Records Chronologically

UC-HIS-03 Categorize Service Records

Use Case Item	Description
Use Case Name	Categorize Service Records
Related Specific Objective	3.3
Primary Actor	Vehicle Owner
Precondition	The selected registered vehicle has saved service records with available service details.
Trigger	The system receives or displays saved service records containing service fields.
Main Flow	The system categorizes saved records using available service record fields such as service type, parts replaced, cost, and shop/mechanic details.
Postcondition	The records can be searched or filtered based on available category fields.

Alternative Flow	If some category details are incomplete, the system categorizes the record using the available fields.
Exception Flow	If no category details are available, the system keeps the record in the service history but marks it as uncategorized.
Requirement ID	FR-HIS-03

Use Case Diagram - UC-HIS-03

Unified Vehicle Service History Consolidation

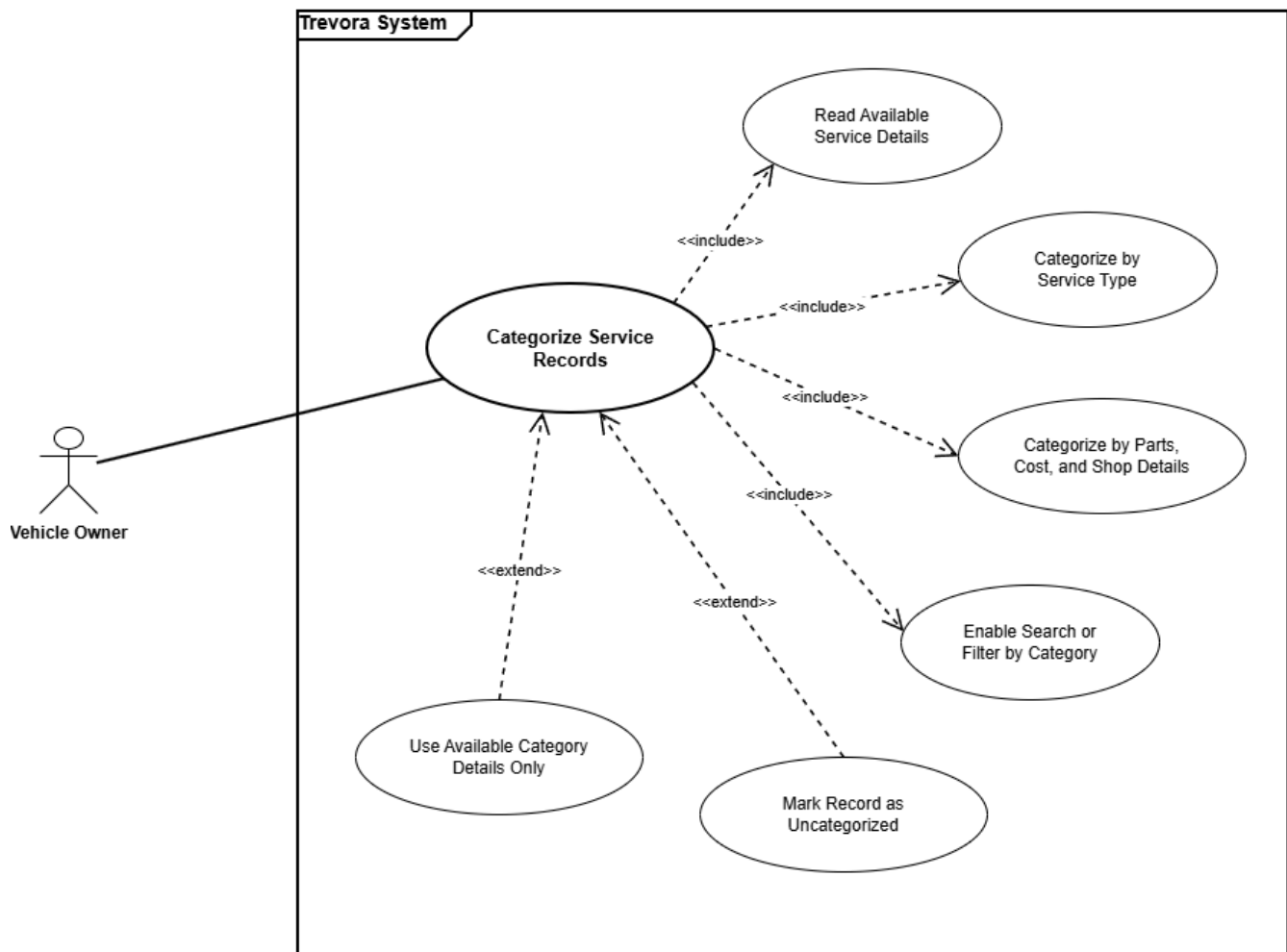


Figure - Use Case Diagram for UC-HIS-03 Categorize Service Records

Activity Diagram - UC-HIS-03
Categorize Service Records

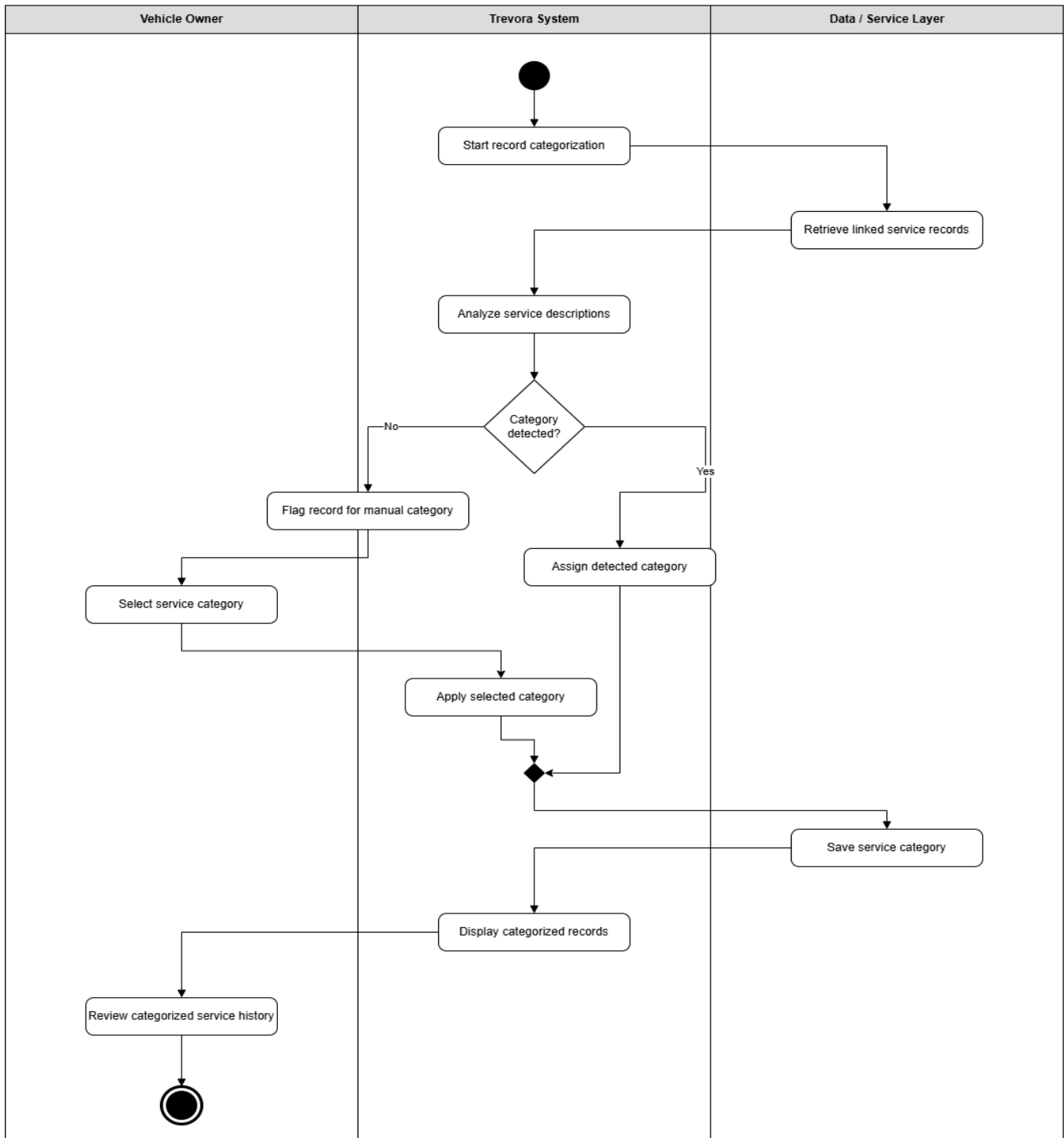


Figure - Activity Diagram for UC-HIS-03 Categorize Service Records

Service History

[+ Add Record]

Toyota Vios 2021 · 12 records

Search service type, shop, parts...

[Filters]

[List]

[Timeline]

[Show Empty State]

Toggle buttons switch between List and Timeline views; Filters button shows/hides filter panel

Status

Source

Sort by

[Select]

[Select]

[Select]

[Clear Filters]

Date	Service Type	Category	Shop	Odometer	Cost	Source	Status
05/07/2026	Oil Change	Maintenance	Superior Auto	62,400 km	₱7,850	Receipt	Verified
03/15/2026	Brake Pad Replacement	Repair	Superior Auto	61,200 km	₱3,200	Manual	Owner
01/02/2026	PMS 40,000 km	Maintenance	Toyota BGC	60,000 km	₱12,500	Receipt	Verified
10/18/2025	Tire Rotation	Maintenance	Giga Tires	58,300 km	₱800	Voice	AI Draft
08/05/2025	Air Filter Replacement	Maintenance	DIY	56,700 km	₱450	Manual	Owner

Clicking a row navigates to Record Detail

Active view: list · filters visible

Figure - Low-Fidelity Wireframe for UC-HIS-03 Categorize Service Records

UC-HIS-04 View Unified Vehicle Service History

Use Case Item	Description
Use Case Name	View Unified Vehicle Service History
Related Specific Objective	3.4
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has selected a registered vehicle profile.
Trigger	The owner opens the service history page of the selected vehicle.
Main Flow	The system displays the complete validated maintenance and repair history of the selected registered vehicle in one centralized service history.
Postcondition	The owner can view the vehicle's validated service records in one place.

Alternative Flow	If no records exist, the system shows an empty history message.
Exception Flow	If the service history cannot be loaded, the system displays an error message and allows the owner to retry.
Requirement ID	FR-HIS-04

Use Case Diagram - UC-HIS-04

Unified Vehicle Service History Consolidation

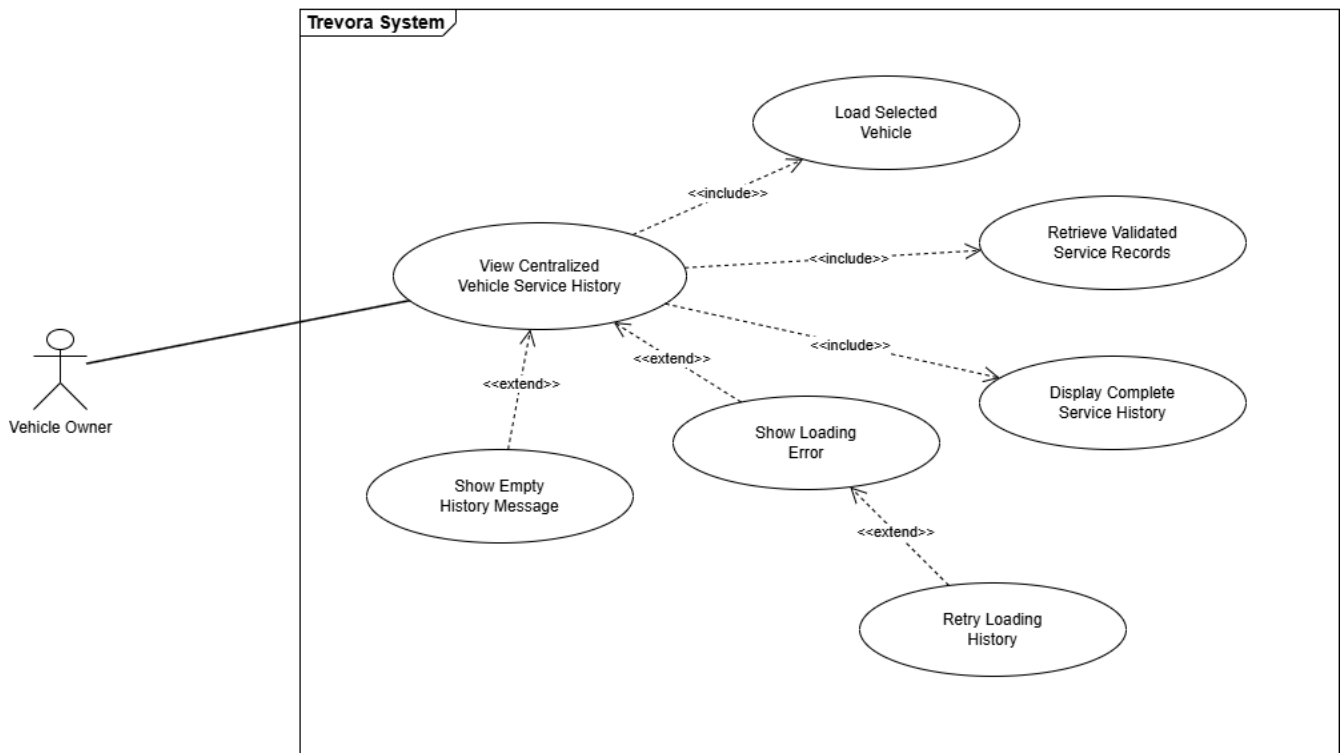


Figure - Use Case Diagram for UC-HIS-04 View Unified Vehicle Service History

Activity Diagram - UC-HIS-04

View Unified Vehicle Service History

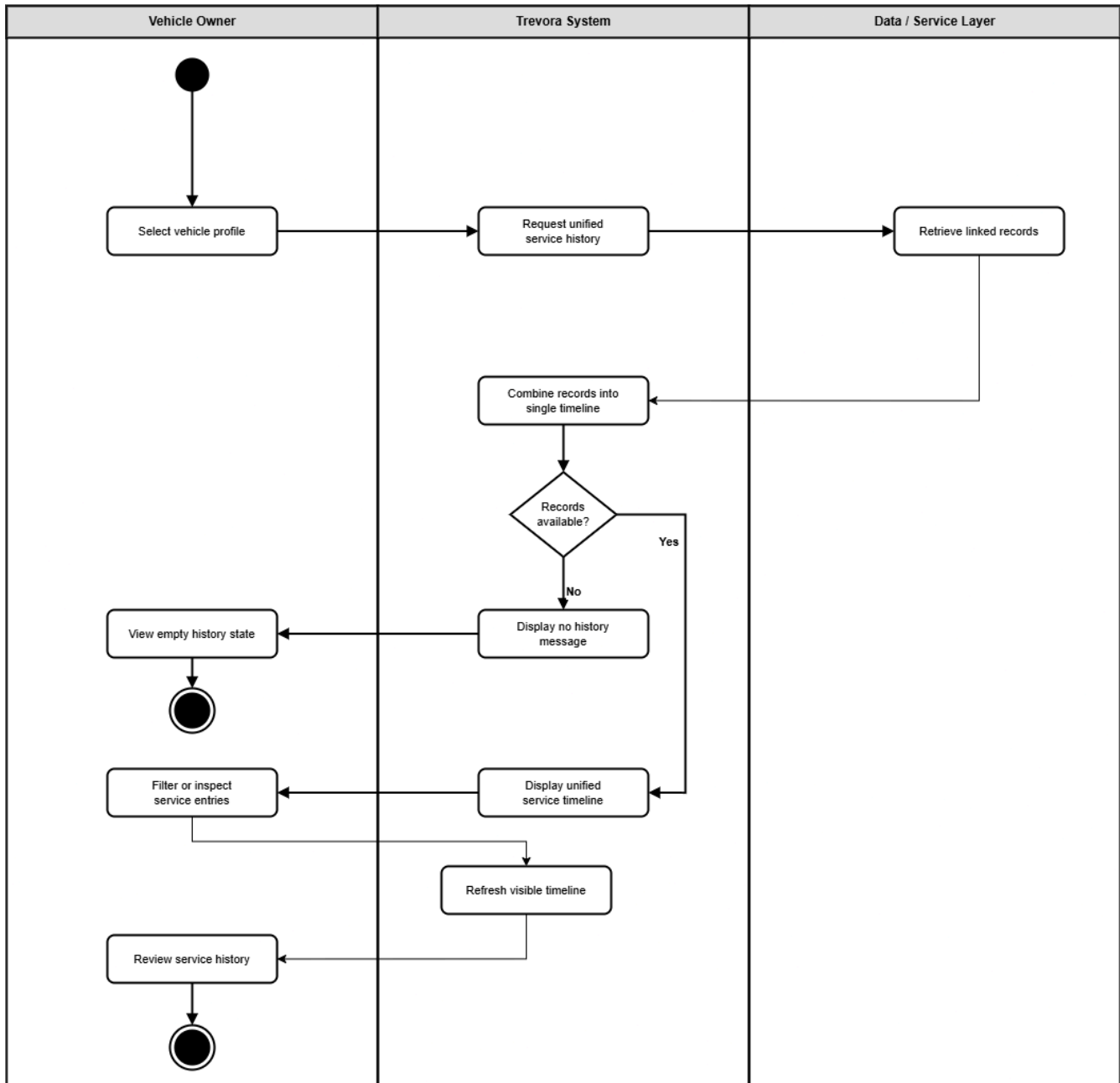


Figure - Activity Diagram for UC-HIS-04 View Unified Vehicle Service History

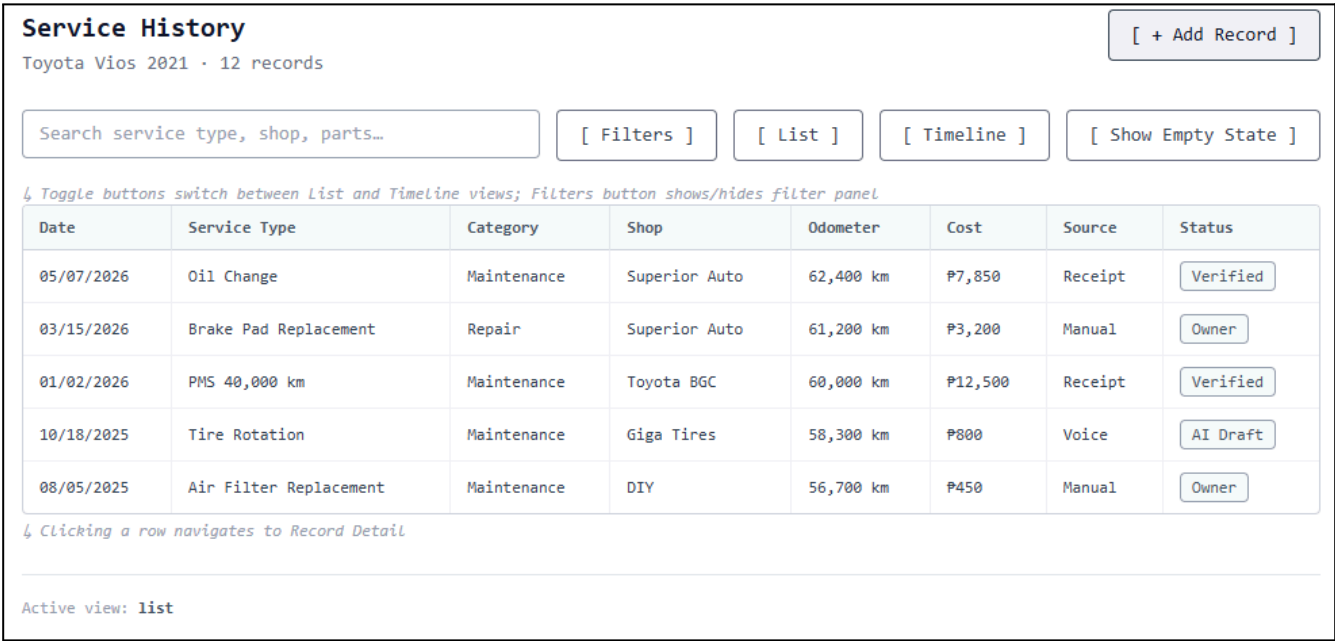


Figure - Low-Fidelity Wireframe for UC-HIS-04 View Unified Vehicle Service History

Module 4: AI-Assisted Service Understanding and Mechanic Handoff

General Objective 4: Improve the use of vehicle service history by providing AI-generated explanations for vehicle owners and QR-based AI record search for mechanics, reducing the time needed to find or understand relevant service information by at least 30%.

Specific Objective-to-Requirement Mapping

Specific Objective	Use Case / Function	Requirement ID
4.1	UC-HAN-01 Show AI-Generated Service Explanation	FR-HAN-01
4.2	UC-HAN-02 Generate QR Access Request	FR-HAN-02
4.3	UC-HAN-03 Notify Temporary Access Expiration	FR-HAN-03
4.4	UC-HAN-04 Approve Mechanic Access Request	FR-HAN-04
4.5	UC-HAN-05 Approve Mechanic Access Request	FR-HAN-05

4.6	UC-HAN-06 Search Shared Records with AI Assistance	FR-HAN-06
Requirement ID	Requirement Statement	
FR-HAN-01	The system shall show AI-generated explanations for selected service records to improve owner understanding.	
FR-HAN-02	The system shall generate a one-time QR access request for the service history of the selected registered vehicle within 5 seconds.	
FR-HAN-03	The system shall notify the vehicle owner that mechanic access is temporary and will expire after a system-set time limit.	
FR-HAN-04	The system shall require owner approval before a mechanic can access the shared vehicle service history.	
FR-HAN-05	The system shall provide temporary read-only mechanic access that automatically expires after a system-set time limit.	
FR-HAN-06	The system shall provide AI-assisted record search for mechanics using only the approved shared vehicle service history.	

UC-HAN-01 Show AI-Generated Service Explanation

Use Case Item	Description
Use Case Name	Show AI-Generated Service Explanation
Related Specific Objective	4.1
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has saved validated service records.
Trigger	The owner opens the service history or views a validated service record.
Main Flow	The system displays an AI-generated explanation for the selected service record, including what was done, why it matters, and what the owner should watch for when available.
Postcondition	The owner can view the original service record and a simple explanation of what was done and why it matters.
Alternative Flow	If an explanation is not yet available, the system generates it when the record is displayed.
Exception Flow	If the AI service is unavailable, the system still shows the original service record and displays a message that the explanation is temporarily unavailable.
Requirement ID	FR-HAN-01

UC-HAN-01: Show AI-Generated Service Explanation

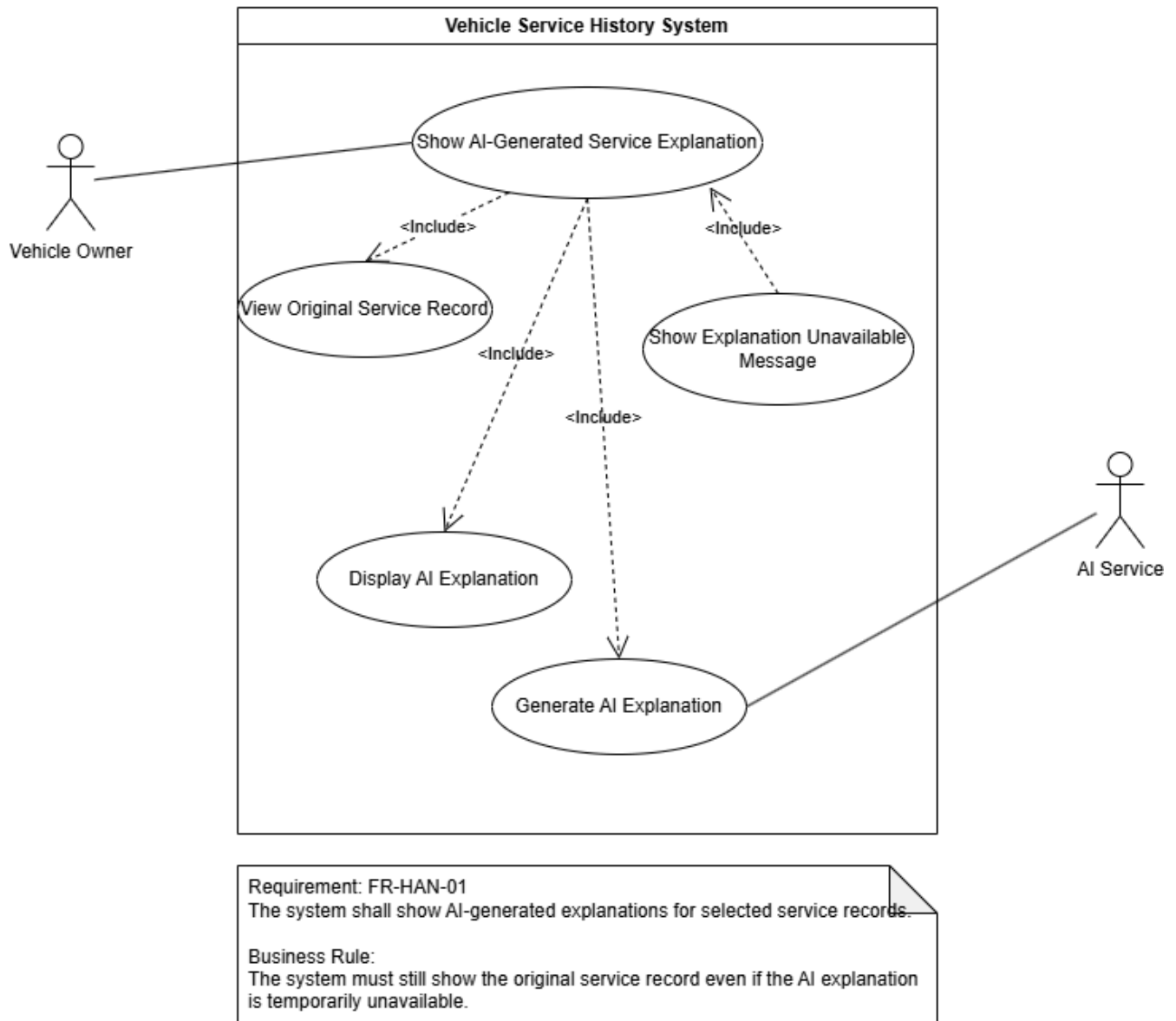


Figure - Use Case Diagram for UC-HAN-01 Show AI-Generated Service Explanation

Activity Diagram - UC-HAN-01
Show AI-Generated Service Explanation

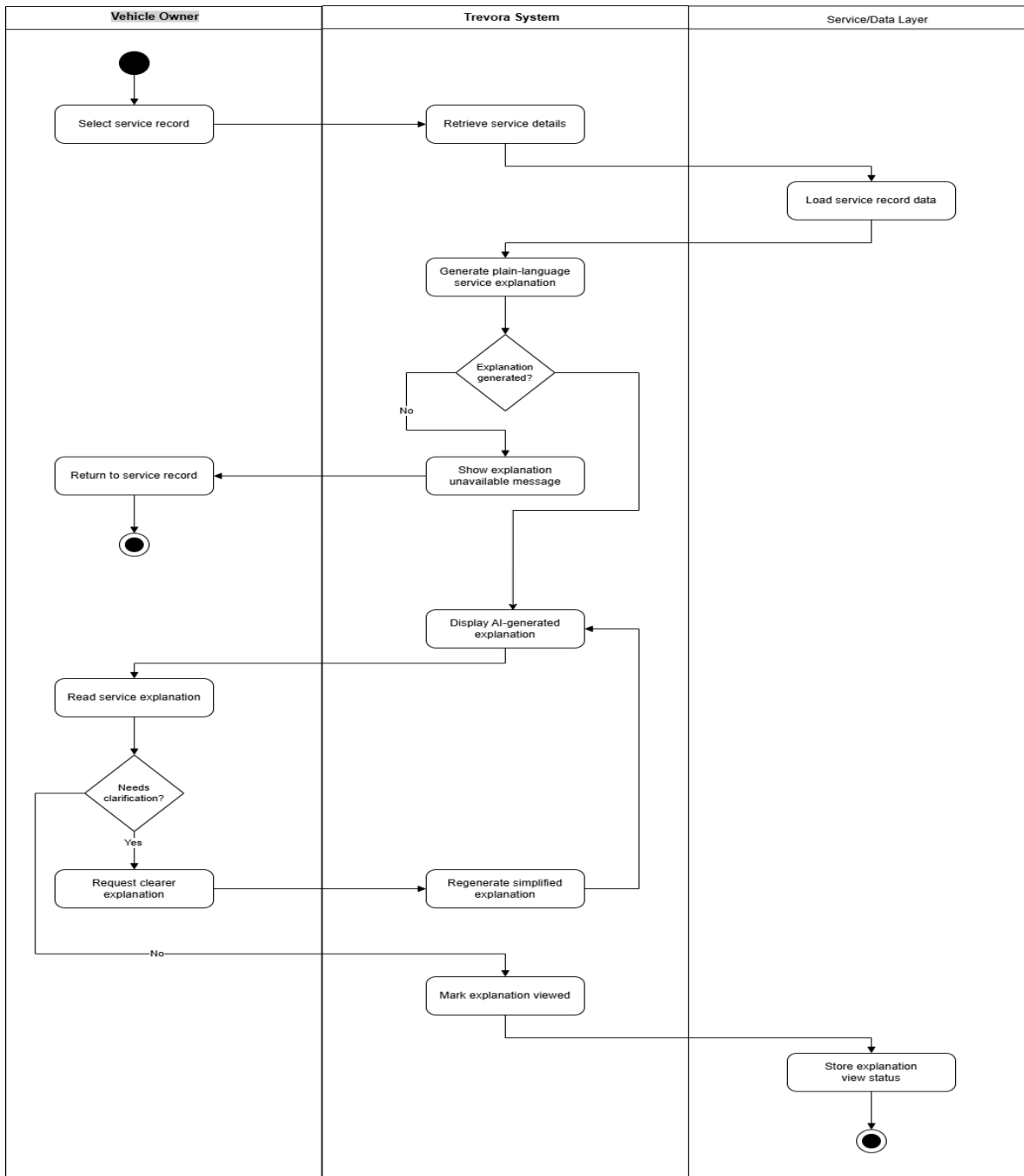


Figure - Activity Diagram for UC-HAN-01 Show AI-Generated Service Explanation

← Back to History

Toyota Vios 2021 · May 7, 2026

✓ Validated

Receipt + AI

[Edit]

[Delete]

Field confidence:

✓ Verified: 1

✓ High: 4

~ Medium: 2

! Low: 2

? Not found: 2

↳ Confidence bar summarises AI extraction quality across all fields

record-details

Record Details

From: Receipt OCR + AI

Vehicle

✓ Verified

Toyota Vios 2021 · ABC 1234

Service Date

✓ High

May 7, 2026

Odometer at Service

? Not found

Not recorded

Service Type

✓ High

Oil Change + Brake Service

Category

✓ High

Maintenance + Repair

Parts Replaced

~ Medium

Oil filter, Brake pads (front + rear)

Work Performed

! Low

Oil drain and refill (4L synthetic), brake pad installation

Shop / Mechanic

~ Medium

Superior Auto Repairs

AI Explanation — gradient header

[Regenerate AI Explanation]

WHAT WAS DONE

WHY IT MATTERS

WHAT TO WATCH FOR

①

②

③

[⚠ AI disclaimer — not a replacement for professional advice]

↳ AI panel: Regenerate refetches explanation; 3 collapsible sections + disclaimer

Figure - Low-Fidelity Wireframe for UC-HAN-01 Show AI-Generated Service Explanation

UC-HAN-02 Generate QR Access Request

Use Case Item	Description
Use Case Name	Generate QR Access Request
Related Specific Objective	4.2
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is logged in and has selected a registered vehicle with saved service records.
Trigger	The owner chooses to share the service history of the selected registered vehicle with a mechanic.

Main Flow	The system creates a one-time QR access request for the service history of the selected registered vehicle and displays the QR code with expiration information.
Postcondition	A one-time QR access request is displayed for scanning within 5 seconds.
Alternative Flow	If the owner changes the vehicle before generating the QR code, the system generates the QR access request for the newly selected vehicle.
Exception Flow	If QR generation fails, the system displays an error message and allows the owner to try again.
Requirement ID	FR-HAN-02

UC-HAN-02: Generate QR Access Request

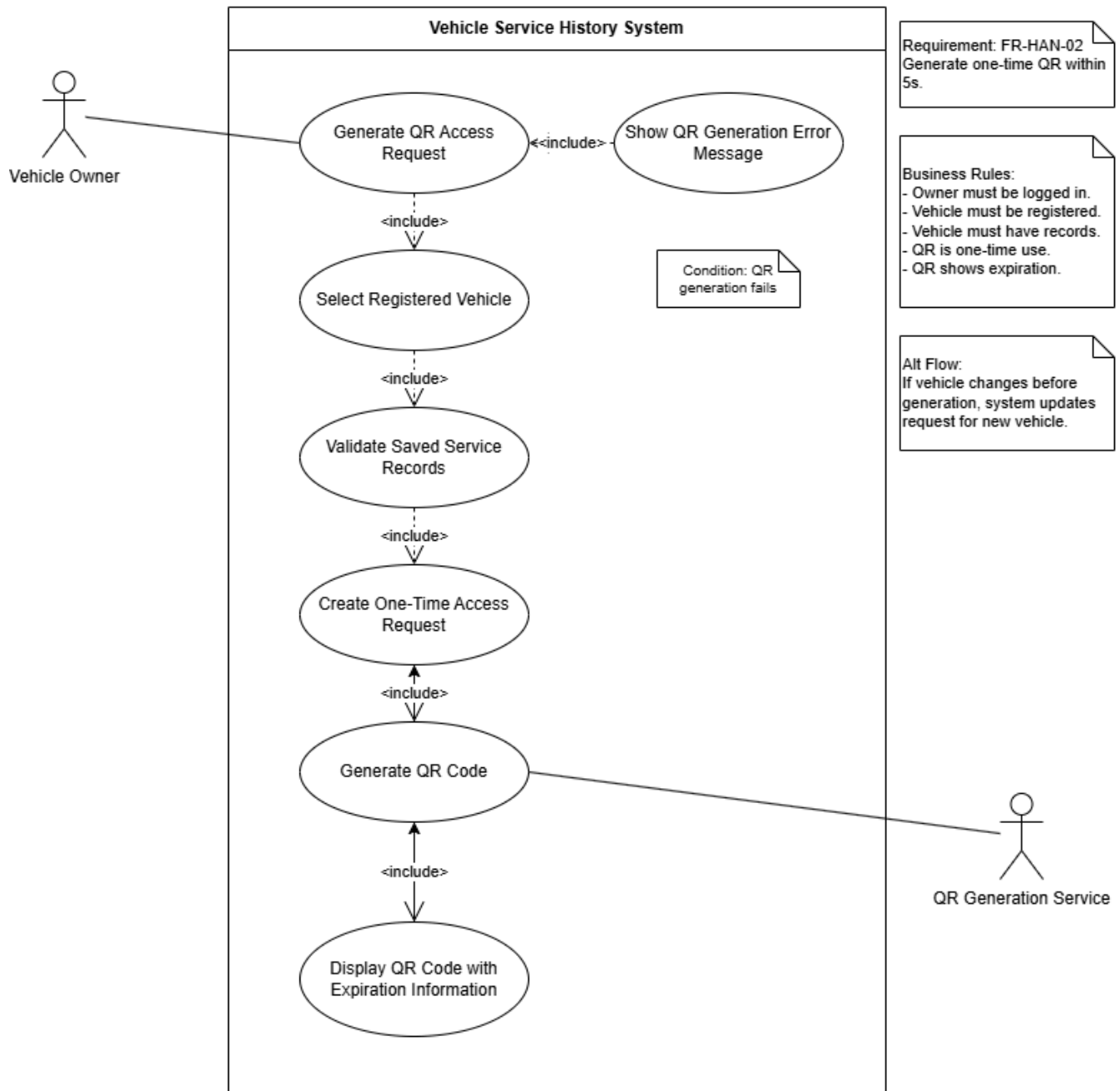


Figure - Use Case Diagram for UC-HAN-02 Generate QR Access Request

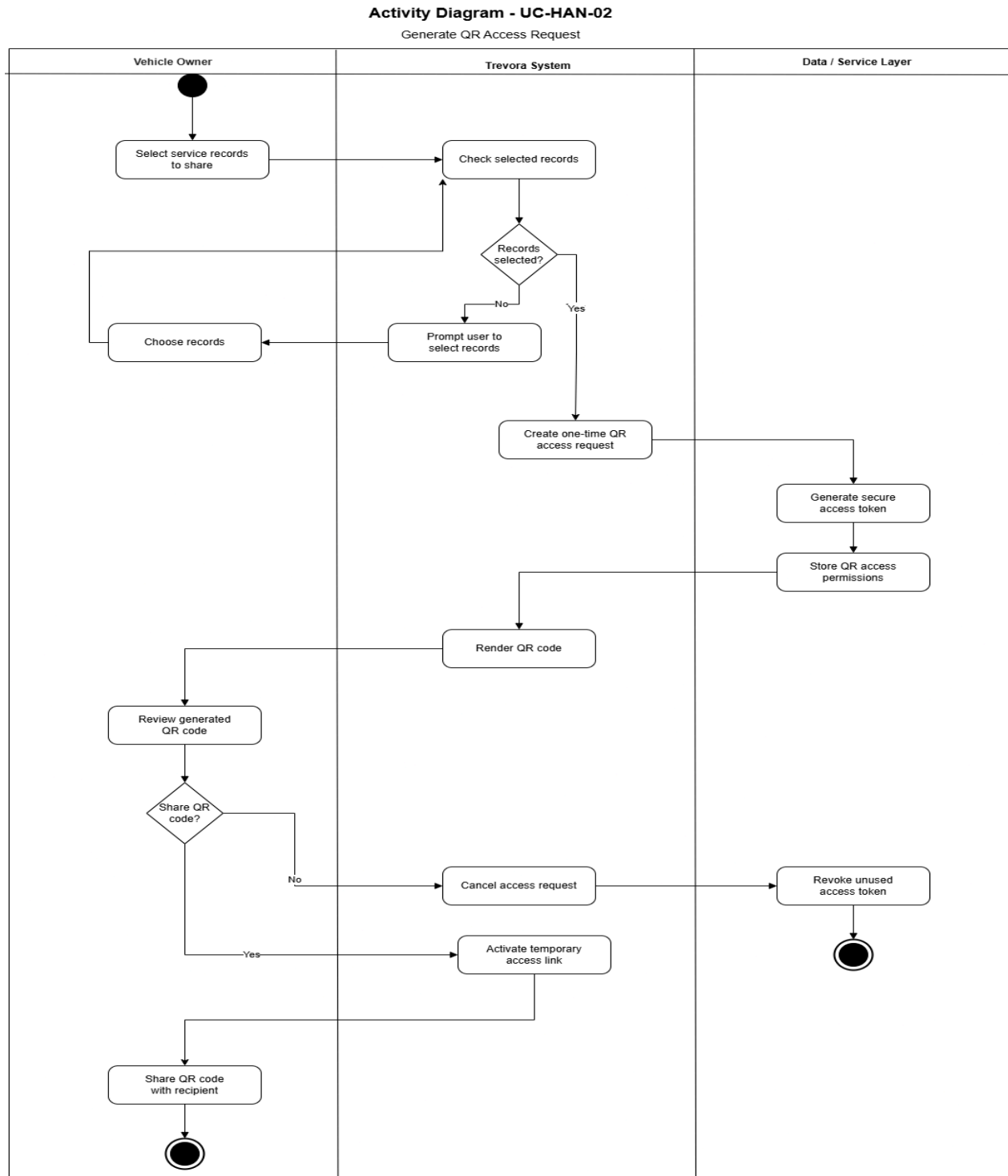


Figure - Activity Diagram for UC-HAN-02 Generate QR Access Request

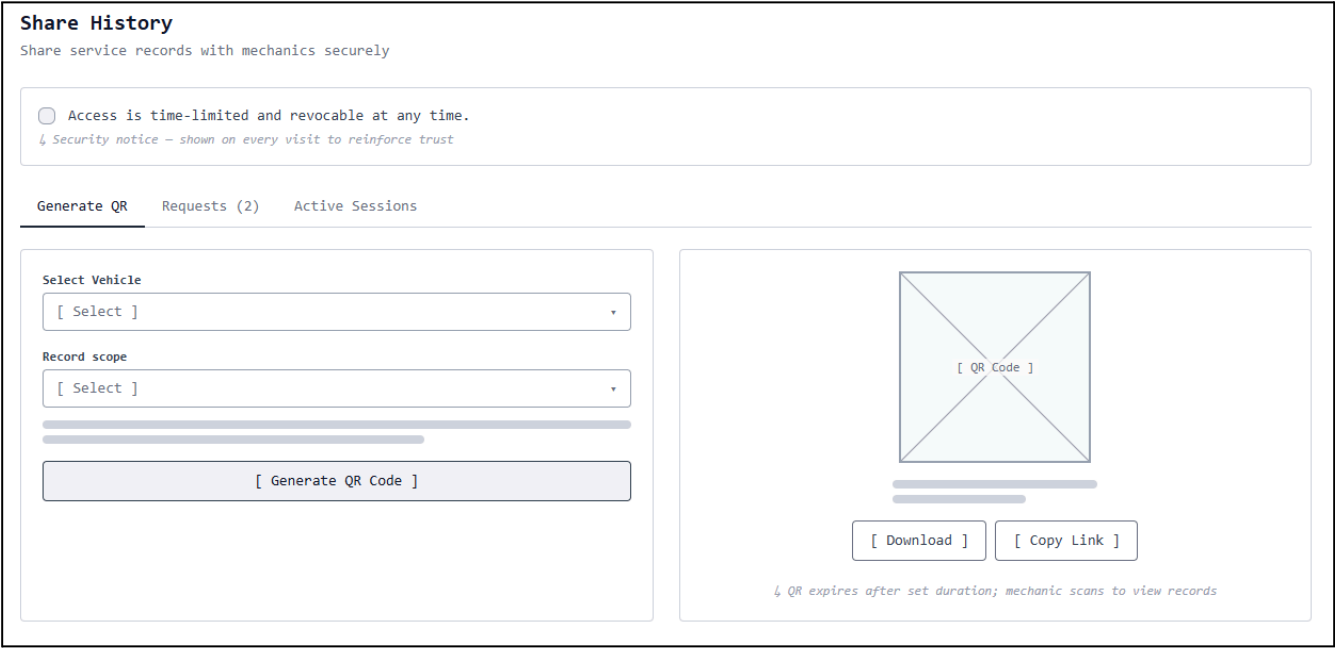


Figure - Low-Fidelity Wireframe for UC-HAN-02 Generate QR Access Request

UC-HAN-03 Notify Temporary Access Expiration

Use Case Item	Description
Use Case Name	Notify Temporary Access Expiration
Related Specific Objective	4.3
Primary Actor	Vehicle Owner
Precondition	The vehicle owner is generating or managing a QR access request for the selected registered vehicle.
Trigger	The owner opens the QR sharing screen or creates a QR access request.
Main Flow	The system displays a notice informing the owner that mechanic access is temporary, read-only, and will expire after a system-set time limit. The system shows the access expiration information before or after the QR access request is generated.
Postcondition	The owner is informed that shared mechanic access is temporary and time-limited.
Alternative Flow	If the owner changes the selected vehicle or sharing scope, the system updates the expiration notice for the current QR access request.

Exception Flow	If expiration information cannot be loaded, the system displays a general notice that mechanic access is temporary and controlled by the system.
Requirement ID	FR-HAN-03

Use Case Diagram - UC-HAN-03

Notify Temporary Access Expiration

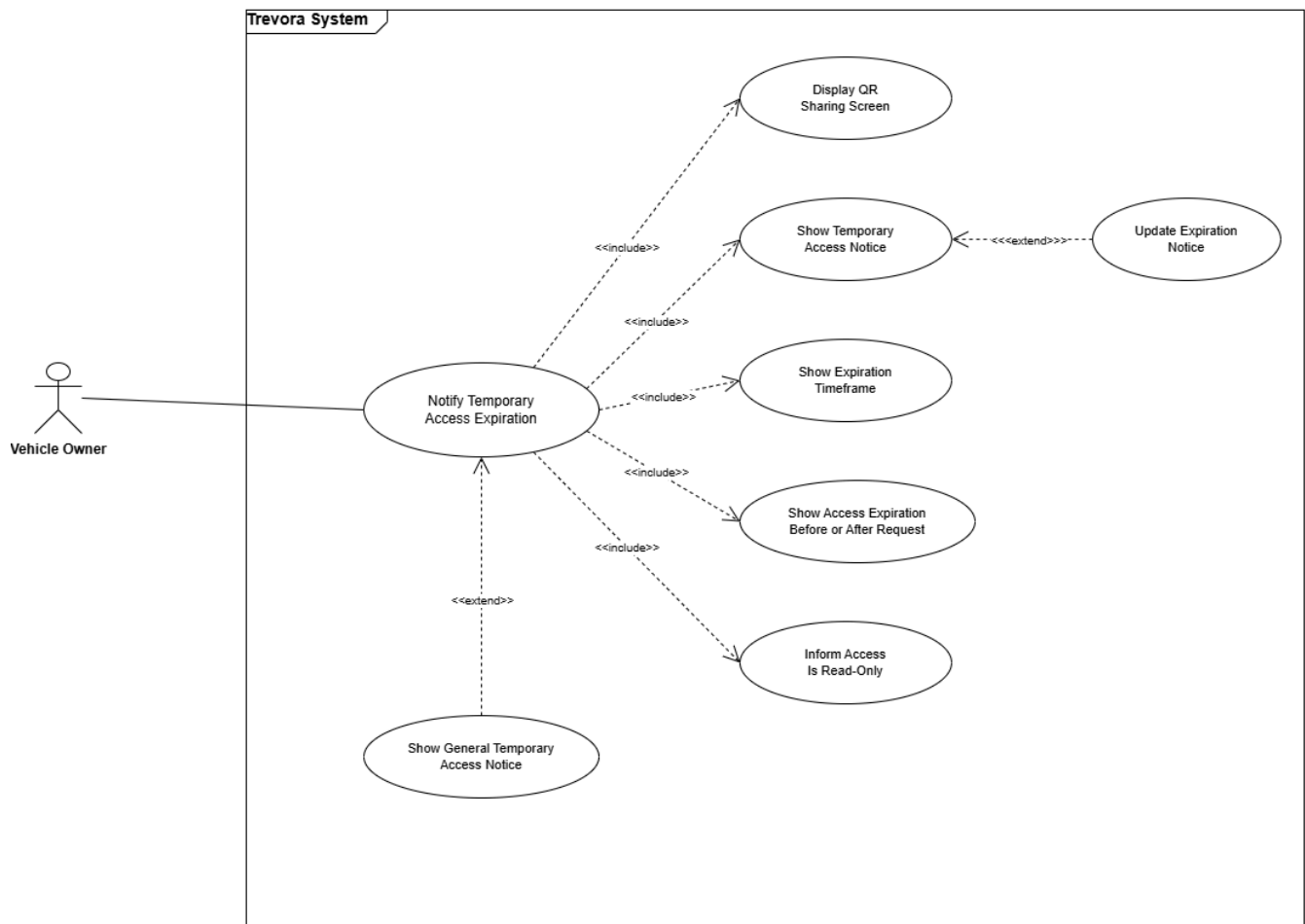


Figure - Use Case Diagram for UC-HAN-03 Notify Temporary Access Expiration

Figure - Activity Diagram for UC-HAN-03 Notify Temporary Access Expiration



Figure - Low-Fidelity Wireframe for UC-HAN-03 Notify Temporary Access Expiration

UC-HAN-04 Approve Mechanic Access Request

Use Case Item	Description
Use Case Name	Approve Mechanic Access Request
Related Specific Objective	4.4
Primary Actor	Vehicle Owner
Secondary Actor	Mechanic / Service Personnel
Precondition	A one-time QR access request has been generated for the selected registered vehicle.
Trigger	The mechanic scans the QR access request and submits an access request.
Main Flow	The system sends the mechanic access request to the vehicle owner. The owner reviews the request and approves mechanic access.
Postcondition	Mechanic access is granted only after owner approval.
Alternative Flow	If the owner denies or ignores the request, mechanic access is not granted.

Exception Flow	If the QR access request is expired or already used, the system blocks the request and asks the owner to generate a new QR code.
Requirement ID	FR-HAN-04

Use Case Diagram - UC-HAN-04

Approve Mechanic Access Request

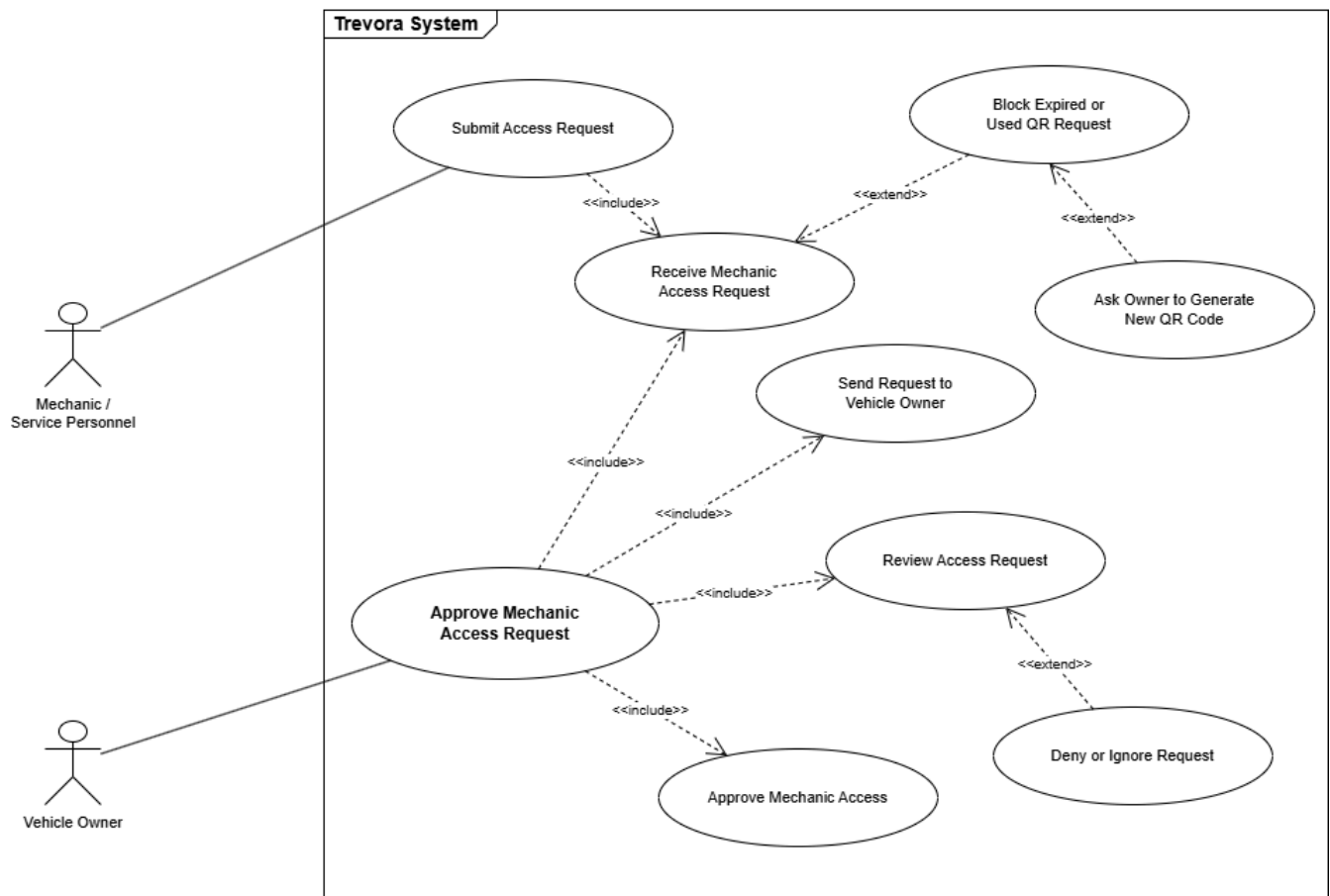


Figure - Use Case Diagram for UC-HAN-04 Approve Mechanic Access Request

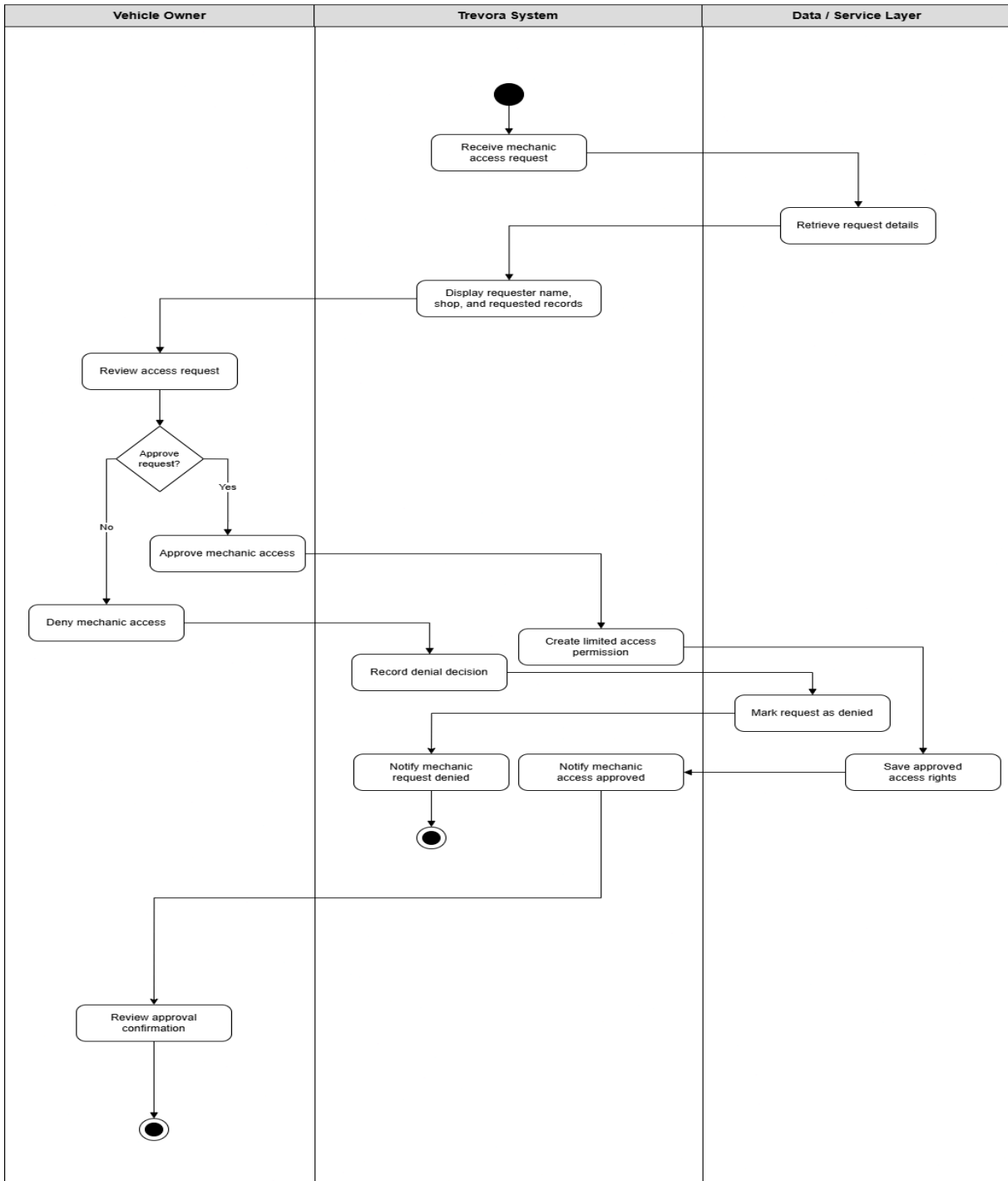


Figure - Activity Diagram for UC-HAN-04 Approve Mechanic Access Request

Share History

Share service records with mechanics securely

☐ Access is time-limited and revocable at any time.
↳ Security notice – shown on every visit to reinforce trust

Generate QR
Requests (2)
Active Sessions

Pending

[Approve]

[Deny]

↳ Mechanic access requests from QR scan

Pending

[Approve]

[Deny]

↳ Mechanic access requests from QR scan

Pending

[Approve]

[Deny]

↳ Mechanic access requests from QR scan

Figure - Low-Fidelity Wireframe for UC-HAN-04 Approve Mechanic Access Request

UC-HAN-05 Provide Temporary Read-Only Access

Use Case Item	Description
Use Case Name	Provide Temporary Read-Only Access
Related Specific Objective	4.5
Primary Actor	Mechanic / Service Personnel
Precondition	The vehicle owner has approved the mechanic access request.
Trigger	The mechanic opens the approved temporary access page.
Main Flow	The system displays the approved vehicle service history in read-only mode until the system-set time limit expires.
Postcondition	The mechanic can view the approved vehicle service history but cannot edit or delete any record.
Alternative Flow	If the mechanic refreshes or reopens the page while access is still active, the system continues showing the approved read-only service history.
Exception Flow	If the access period expires, the system blocks further viewing and displays a message that access has expired.

Requirement ID	FR-HAN-05
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Use Case Diagram - UC-HAN-05

Provide Temporary Read-Only Access

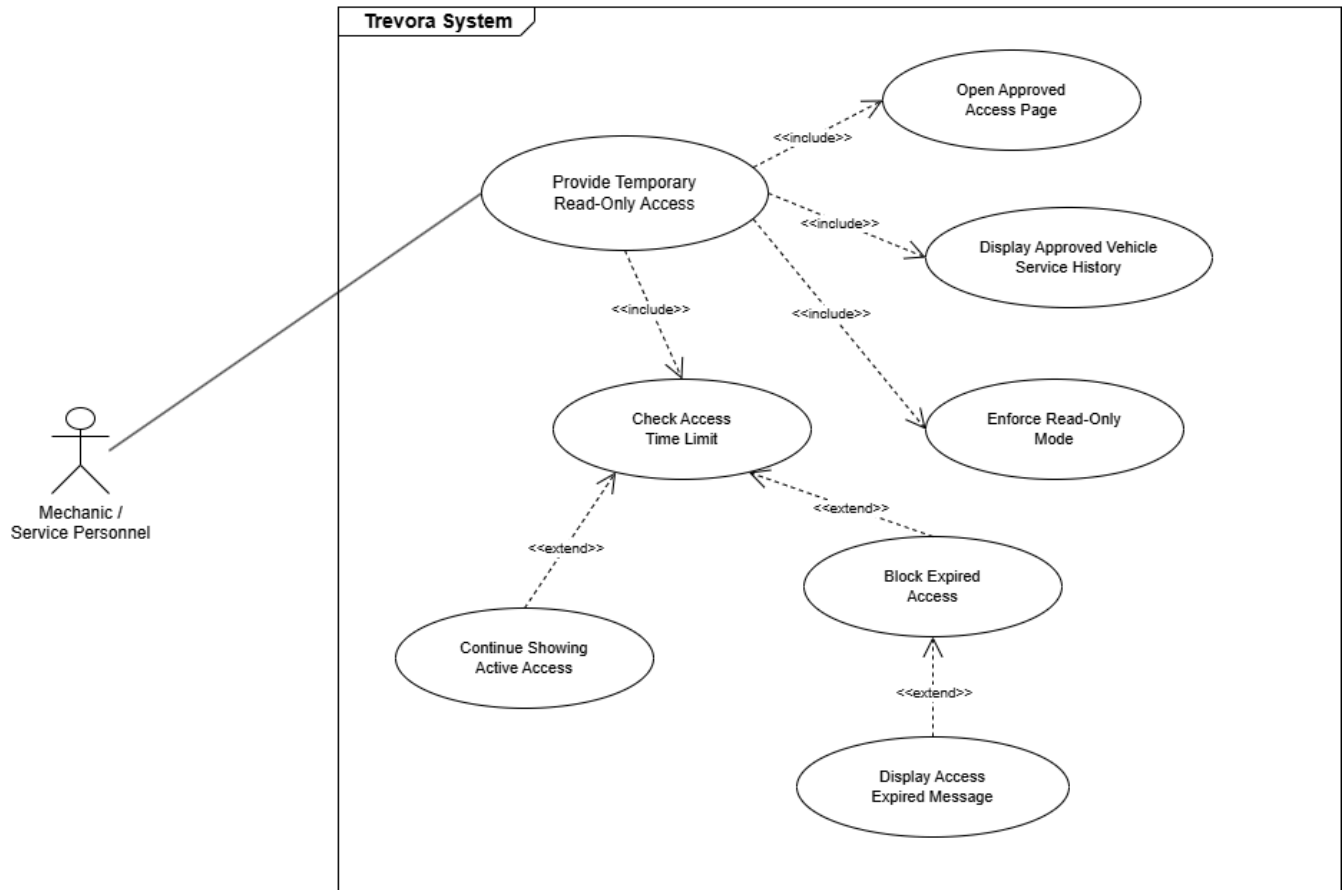


Figure - Use Case Diagram for UC-HAN-05 Provide Temporary Read-Only Access

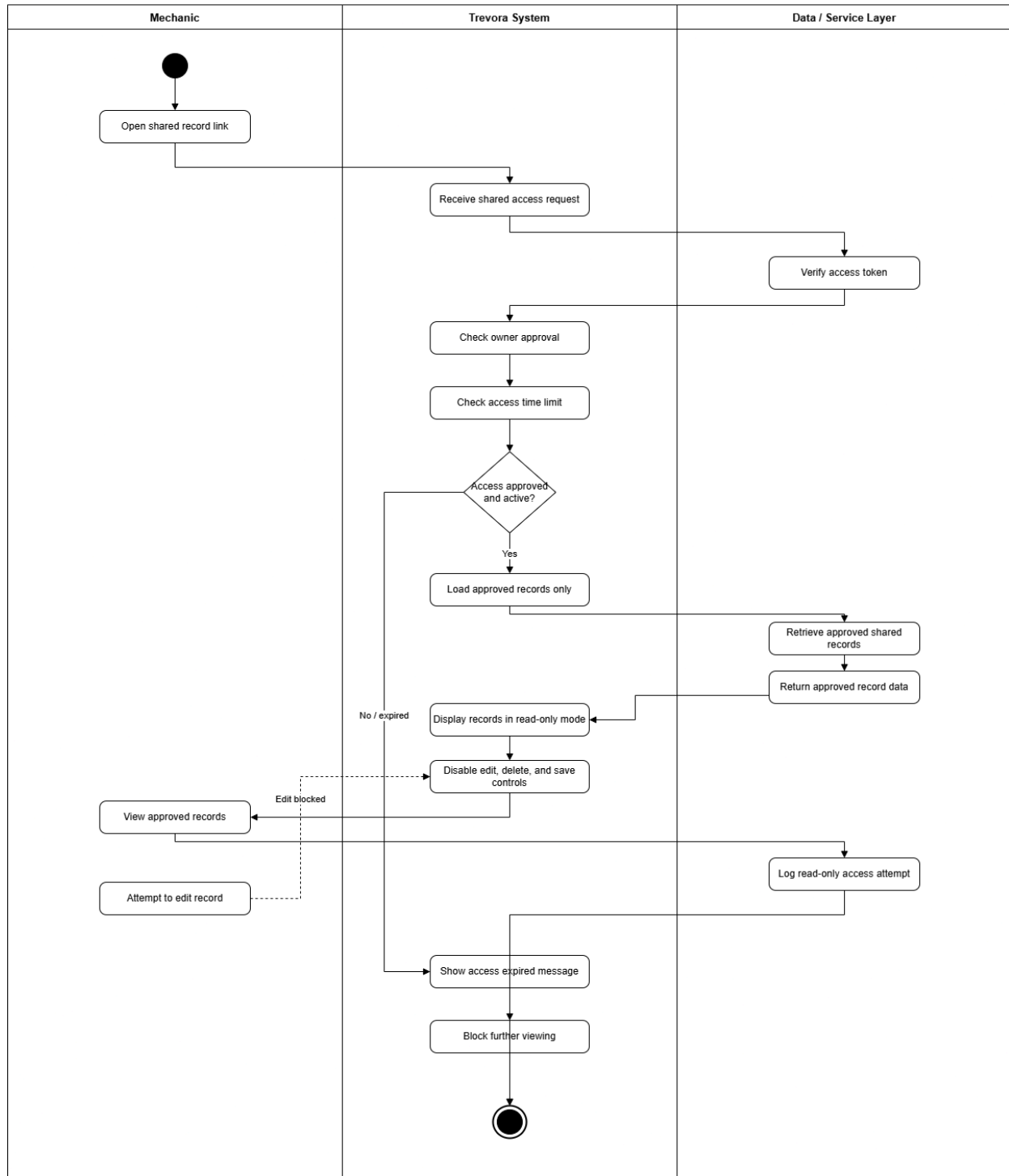


Figure - Activity Diagram for UC-HAN-05 Provide Temporary Read-Only Access

access-notice

Access active

Scope: All records5 recordsRead-only

↳ Access notice: owner name, vehicle, granted scope, record count – no edit controls

AI Query – gradient-header (blue → purple)

[Ask]

↳ AI input: icon-prefixed text field + Ask button; X clear button appears when text present

Last oil change?Brake history?Upcoming maintenance?Total spend?Last PMS date?

↳ Quick query chips: clicking populates input and auto-submits

AI Response

Related records: r1

↳ Response area: AI answer text + highlighted record chips (clicking scrolls to record)

Service Records5 records

May 7, 2026

ReceiptValidated

✚ AI match

[View]

Mar 2, 2026

VoiceValidated

[View]

Jan 15, 2026

ManualPending

[View]

Figure - Low-Fidelity Wireframe for UC-HAN-05 Provide Temporary Read-Only Access

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UC-HAN-06 Search Shared Records with AI Assistance

Use Case Item	Description
Use Case Name	Search Shared Records with AI Assistance
Related Specific Objective	4.6
Primary Actor	Mechanic / Service Personnel
Precondition	Temporary read-only access is active and the mechanic is viewing the approved vehicle service history.
Trigger	The mechanic enters a service-related question or keyword in the AI-assisted search field.
Main Flow	The system searches the approved shared vehicle service history and returns related service records based on the mechanic's input.
Postcondition	Related service records are displayed without exposing records from other vehicles or unapproved accounts.
Alternative Flow	If no related record is found, the system shows an empty result message.
Exception Flow	If the AI-assisted search is unavailable, the system displays an error message and allows the mechanic to manually browse the approved service history.
Requirement ID	FR-HAN-06

Use Case Diagram - UC-HAN-06

Search Shared Records with AI Assistance

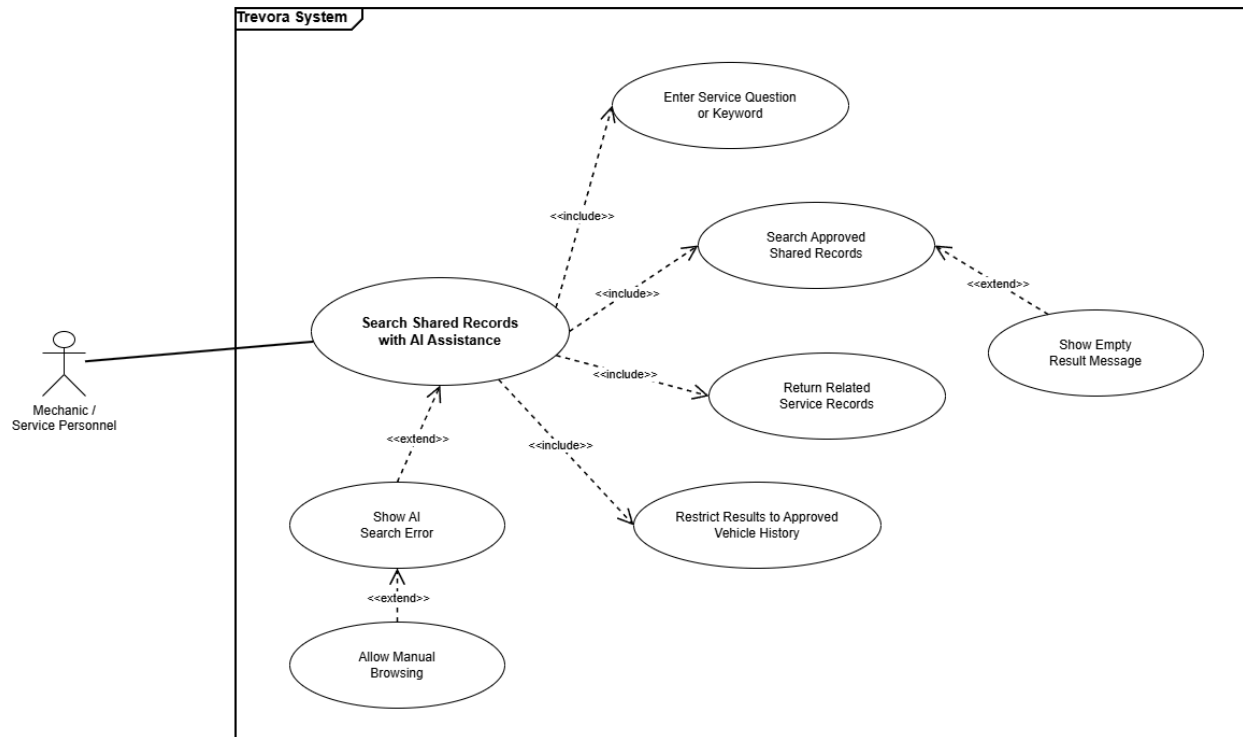


Figure - Use Case Diagram for UC-HAN-06 Search Shared Records with AI Assistance

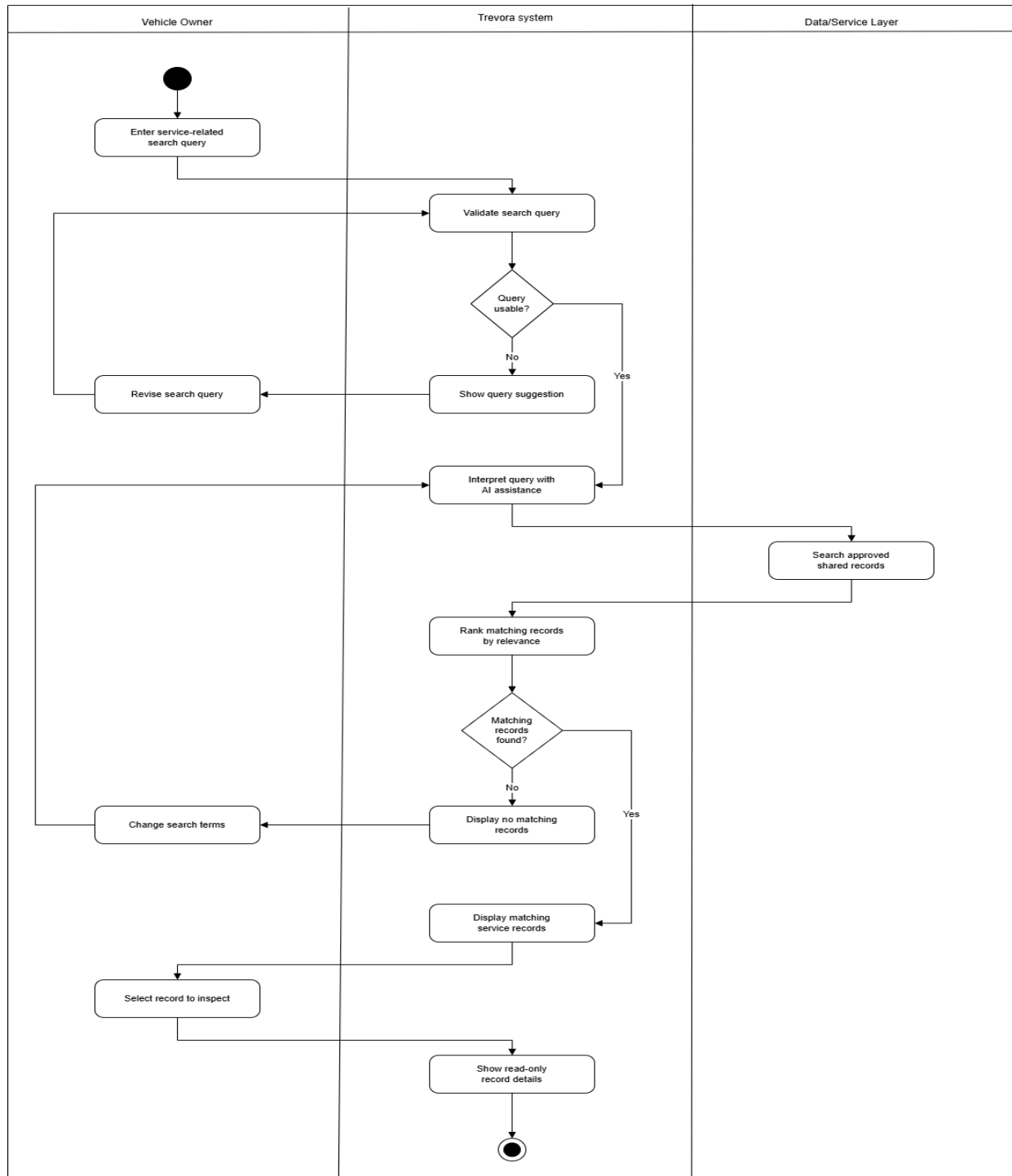


Figure - Activity Diagram for UC-HAN-06 Search Shared Records with AI Assistance

↳ Access notice: owner name, vehicle, granted scope, record count – no edit controls

AI Query – gradient header (blue → purple)

[Ask]

↳ AI input: icon-prefixed text field + Ask button; X clear button appears when text present

Last oil change?

Brake history?

Upcoming maintenance?

Total spend?

Last PMS date?

↳ Quick query chips: clicking populates input and auto-submits

AI Response

Related records:

r1

↳ Response area: AI answer text + highlighted record chips (clicking scrolls to record)

Figure - Low-Fidelity Wireframe for UC-HAN-06 Search Shared Records with AI Assistance

Non-functional requirements

Performance

NFR-PER-01: The system shall load the service data entry screen within 3 seconds for at least 90% of test attempts under normal internet conditions.

NFR-PER-02: The system shall save a validated service record within 5 seconds after user confirmation, excluding third-party service outages.

NFR-PER-03: The system shall display the first 100 service history records within 5 seconds and support pagination or filtering for larger histories up to 500 records.

NFR-PER-04: The system shall generate a one-time QR access request for the service history of the selected registered vehicle within 5 seconds after the owner submits the request.

NFR-PER-05: The mechanic search feature shall reduce relevant record-finding time by at least 30% in task-based testing compared with manual browsing of the shared vehicle service history.

Security

NFR-SEC-01: The system shall require authenticated access before users can create, view, update, or share service records.

NFR-SEC-02: The system shall enforce role-based permissions for vehicle owners, mechanics, and administrators using server-side access checks.

NFR-SEC-03: The system shall require owner approval before a mechanic can view the shared vehicle service history through a QR access request.

NFR-SEC-04: The system shall provide mechanics with temporary read-only access only; mechanics shall not be allowed to create, edit, or delete owner service records.

NFR-SEC-05: The system shall automatically expire temporary mechanic access after the configured time limit and deny access to vehicle records not approved by the owner.

Reliability

NFR-REL-01: The system shall preserve saved validated service records unless the owner or an authorized administrator performs a permitted delete action.

NFR-REL-02: The system shall prevent saving when required fields are missing, including vehicle profile, service date, service type, and cost.

NFR-REL-03: The system shall reduce incorrect or incomplete saved entries by at least 30% through review, validation, correction, and confirmation steps.

NFR-REL-04: The system shall associate each saved record with the selected vehicle profile and verify ownership before storing the record.

NFR-REL-05: The system shall provide manual entry, retry, or editable review options when OCR, speech-to-text, or AI-assisted services return incomplete results.